From Apportionment to Boundary Design: Extending Huntington-Hill to Congressional Redistricting

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Abstract

For over eighty years, the Huntington-Hill method has governed congressional apportionment—determining how many seats each state receives—through transparent mathematical formulas that command universal acceptance. Yet the parallel problem of redistricting—drawing district boundaries within states—remains mired in partisan conflict, with the Supreme Court declaring such disputes non-justiciable in *Rucho v. Common Cause* (2019). We ask: if mathematical objectivity resolved the politically charged question of *how many* seats each state receives, why not apply similar principles to *where* district boundaries are drawn? This paper presents a recursive graph bisection algorithm for congressional redistricting that extends Huntington-Hill's commitment to mathematical rigor from apportionment to boundary design. Applied to all 50 states using 2020 Census data, the algorithm produces 435 congressional districts with mean population deviation of 2.79%, guaranteed geographic contiguity, and structural immunity to partisan manipulation—the algorithm cannot access voter registration, election results, or any partisan data. We analyze population balance, compactness, political characteristics, and demographic representation, finding that algorithmic redistricting can satisfy traditional criteria while eliminating direct opportunities for gerrymandering. Our results suggest that mathematical governance offers a viable path toward resolving redistricting conflicts and restoring public confidence in electoral fairness.

Keywords

Redistricting, Gerrymandering, Graph Partitioning, Congressional Apportionment, Electoral Reform, Huntington-Hill Method

1 Introduction

Every decade, American states redraw congressional district boundaries to reflect population changes revealed by the decennial census. This redistricting process determines political representation for over 330 million Americans across 435 House seats. Yet despite its profound impact on democratic governance, redistricting remains one of the most contentious and manipulated aspects of American electoral politics.

The problem is not new. Partisan gerrymandering—the strategic manipulation of district boundaries to advantage one party over another—has plagued American democracy since Elbridge Gerry's salamander-shaped Massachusetts district in 1812. What has changed is the sophistication of the manipulation. Modern computing power enables unprecedented precision in identifying partisan voters and crafting districts to maximize electoral advantage **?**. Gerrymandered maps can virtually guarantee election outcomes before a single vote is cast, transforming the fundamental principle of representative democracy—that voters choose their representatives—into its inverse: representatives choosing their voters.

The Supreme Court's 2019 decision in *Rucho v. Common Cause* **?** effectively removed federal judicial oversight of partisan gerrymandering, declaring such disputes non-justiciable political questions beyond the reach of federal courts. This ruling left redistricting primarily in the hands of state legislatures, where partisan incentives often dominate. The result is predictable: in states where one party controls the redistricting process, maps tend to favor that party. Public confidence in electoral fairness has eroded accordingly.

Reform efforts have focused largely on institutional solutions: independent redistricting commissions, nonpartisan staff agencies, and enhanced transparency requirements. These approaches represent important progress. Yet they share a fundamental limitation—they replace one set of human decision-makers with another, substituting partisan operatives with "independent" commissioners. The humans change, but human judgment remains central. Commissioners must still decide where to draw lines, how to balance competing criteria, and which communities to unite or divide. Even well-intentioned humans possess political knowledge that inevitably influences their decisions, consciously or otherwise.

This paper explores a different approach: removing human judgment from boundary-drawing decisions entirely through algorithmic redistricting. The algorithm we present uses only census data—population counts and geographic adjacency—to produce congressional district maps. It cannot access voter registration, election results, partisan affiliation, or any political information. The boundaries it draws reflect population distribution and geographic optimization, not strategic calculation or backroom negotiation.

1.1 The Huntington-Hill Precedent

Our proposal rests on a powerful precedent: for over 80 years, Congress has successfully used a mathematical formula—the Huntington-Hill method—to allocate House seats among states. This apportionment problem was historically even more contentious than redistricting, with Congress changing formulas after nearly every census between 1790 and 1940 to favor particular states or political coalitions **?**. The instability ended in 1941 when Congress adopted the Huntington-Hill method **?**, which has operated without controversy ever since.

The method's success stems not from optimal outcomes—mathematical proofs show that no apportionment method satisfies all reasonable fairness criteria simultaneously **?**—but from procedural legitimacy. The Huntington-Hill formula is transparent, reproducible, and treats all states identically. Most critically, it removes human discretion from a politically sensitive decision. No state can credibly claim unfair treatment when seat allocation follows from explicit mathematical principles applied uniformly nationwide.

We argue that the same philosophy can govern redistricting. If mathematical objectivity resolved the politically charged question of *how many* seats each state receives, why not apply similar principles to *where* district boundaries should be drawn? Both questions involve dividing political power according to population. Both invite partisan manipulation when left to political actors. Both require balancing competing criteria that cannot all be perfectly satisfied. And both can benefit from mathematical formalization that transforms contentious political disputes into technical implementation questions.

1.2 Our Contribution

This paper presents a recursive graph bisection algorithm for congressional redistricting inspired by the Huntington-Hill precedent. We model each state as a graph where census tracts form nodes and geographic adjacency defines edges. Using the METIS graph partitioning library **?**, we recursively divide this graph to create districts that:

- Achieve near-perfect population equality (mean deviation: 2.79%)
- Maintain geographic contiguity by construction
- Optimize compactness through edge-cut minimization
- Produce reproducible results from the same input data
- Possess structural immunity to partisan manipulation

We apply this method to all 50 states using 2020 Census tract-level population data **?**, creating a complete nationwide congressional district plan. We then overlay 2020 presidential election results **?** onto the algorithmically-generated districts to analyze their political characteristics—examining what partisan patterns emerge when political data plays no role in boundary decisions.

Our analysis reveals several key findings:

Technical feasibility: Recursive bisection successfully handles diverse state characteristics—from Wyoming's single district to California's 52 districts, from compact states to island territories. The algorithm achieves population balance comparable to carefully-drawn human maps while guaranteeing geographic contiguity.

Partisan patterns reflect geography, not manipulation: The resulting districts exhibit a systematic Democratic advantage in district count (56.5% Democratic-leaning) despite the algorithm's complete ignorance of partisan data. This efficiency gap arises from well-documented geographic sorting: urban Democratic voters concentrate in compact areas while rural Republican voters disperse across larger regions **??**. The pattern confirms that compactness-optimizing algorithms cannot eliminate efficiency gaps caused by political geography—but critically, the algorithm cannot intentionally create or amplify such gaps for partisan advantage.

The impossibility defense: Traditional arguments against gerrymandering focus on intent—did mapmakers deliberately manipulate boundaries for partisan gain? This standard proves difficult to apply legally and institutionally, as *Rucho* demonstrated. Our algorithm offers a stronger defense: *impossibility*. The algorithm cannot gerrymander because it cannot see partisan data. It draws boundaries based on population and adjacency alone. Any partisan patterns in the results reflect the interaction between geographic voter distribution and compactness optimization, not strategic design.

Process fairness over outcome fairness: Political scientists and legal scholars debate what constitutes "fair" political representation: proportional representation (seat share matching vote share), competitive districts (maximizing electoral contestability), partisan symmetry (equal treatment of parties), or communities of interest (keeping like-minded voters together). Our algorithm does not optimize for any single fairness metric. Instead, it prioritizes *process fairness*—transparent, reproducible decision-making that treats all geographic units identically and cannot be manipulated to achieve predetermined outcomes. This philosophical stance aligns with Huntington-Hill: accept that political geography constrains outcomes, but ensure the process cannot be gamed.

1.3 Normative Implications

Algorithmic redistricting raises profound questions about democratic governance. When should mathematical formulas make political decisions? What role remains for human judgment when algorithms draw boundaries? How do we balance competing values—transparency versus flexibility, objectivity versus community input, neutrality versus affirmative obligations like Voting Rights Act compliance?

We do not claim algorithms should replace all human judgment in redistricting. Values questions—how much weight to give compactness versus communities of interest, whether to prioritize competitive districts or partisan fairness—require normative choices that algorithms cannot make independently. Algorithms are tools, not oracles.

But we contend that once humans establish redistricting criteria, algorithms can implement those criteria more objectively, transparently, and consistently than any human commission. More fundamentally, algorithmic redistricting addresses the core legitimacy crisis in American redistricting: the widespread belief that district maps serve politicians' interests rather than voters' interests. When an algorithm that cannot see political data draws boundaries, voters can trust that geographic and population considerations—not partisan calculation—determined the outcome.

Eighty years ago, Congress faced an analogous legitimacy crisis in apportionment. Huntington-Hill resolved it not by finding perfect fairness—which mathematically cannot exist—but by establishing procedural legitimacy through mathematical objectivity. Redistricting today faces a similar crisis. Algorithmic approaches offer a similar solution.

1.4 Research Program Context

This paper introduces the foundational method for a broader research program investigating recursive bisection for congressional redistricting. Subsequent work extends this baseline approach with compactness optimization through edge-weighted graph partitioning, adds demographic awareness for Voting Rights Act compliance, and systematically investigates the method's properties across parameter variations, temporal stability over multiple census decades, and comparisons to alternative partitioning strategies. Together, these studies demonstrate that graph-based redistricting offers a computationally efficient, legally defensible, and democratically legitimate approach to boundary design that addresses the core legitimacy crisis identified above.

1.5 Road Map

The remainder of this paper proceeds as follows. Section ?? examines the Huntington-Hill precedent in detail, identifying the principles that enabled mathematical governance of apportionment and how those principles extend to redistricting. Section ?? describes our recursive bisection algorithm, including graph construction, water-based adjacency handling, and the METIS partitioning library. Section ?? presents population balance and compactness statistics for all 435 algorithmically-generated districts. Section ?? overlays election data to analyze political characteristics and partisan patterns. Section ?? discusses advantages and limitations, compares our approach to alternatives, and examines policy implications. Section ?? concludes with reflections on mathematical governance and democratic legitimacy.

2 The Huntington-Hill Precedent

Congressional apportionment—determining how many House seats each state receives—demonstrates that mathematical formulas can successfully govern politically sensitive decisions when properly designed. This section examines why Huntington-Hill succeeded where previous methods failed, and how its underlying principles extend to redistricting.

2.1 The Apportionment Problem and Its Political History

The Constitution requires that House seats be apportioned among states according to population, with each state guaranteed at least one representative. For a fixed House size (435 members since 1913), this creates a discrete allocation problem: states with fractional entitlements must be rounded either up or down, and different rounding rules advantage different states.

From 1790 to 1940, Congress changed apportionment methods after nearly every census, selecting whichever formula benefited the politically dominant coalition ?. Methods varied—Jefferson (1792–1840), Webster (1840–1850, 1910–1930), Hamilton (1850–1900), and others—with each change sparking fierce debate about fairness. States that would lose seats under a new method inevitably claimed discrimination.

This chronic instability demonstrated that *any* apportionment method makes someone unhappy. The mathematical reality, proven rigorously by Balinski and Young ?, is that no apportionment method can simultaneously satisfy all reasonable fairness criteria. Every method embodies normative trade-offs between competing principles: minimizing maximum relative differences versus minimizing average absolute differences, favoring small states versus large states, maintaining proportionality versus avoiding quota violations.

The choice of method necessarily involves value judgments. Yet Congress's repeated method-switching revealed that when political actors make those judgments, the result is perpetual conflict. Each census brought renewed fights as states maneuvered for favorable formulas.

2.2 The 1941 Resolution: Permanent Adoption of Huntington-Hill

In 1941, Congress made a crucial procedural change: it adopted the Huntington-Hill method *permanently*, embedding it in statute rather than choosing it anew each census **?**. This decision transformed apportionment from a recurring political battle into settled policy.

The Huntington-Hill method works as follows. After each state receives its guaranteed seat, remaining seats are allocated iteratively. Each state's priority for receiving its n-th seat is computed as:

$$P_n = \frac{\text{Population}}{\sqrt{n(n-1)}}$$

The state with the highest priority value receives the next seat. This process continues until all 435 seats are allocated.

The geometric mean denominator $\sqrt{n(n-1)}$ minimizes the relative difference in representation between individuals across states. That is, it minimizes the maximum value of $|r_i - r_j| / \max(r_i, r_j)$ where r_i is the representation ratio (population per seat) in state i. This particular fairness criterion has an appealing interpretation: it treats inequalities symmetrically, making an individual in an under-represented state's complaint equally valid to an individual in an over-represented state's complaint.

But Huntington-Hill's success derives not from mathematical optimality—it still violates some fairness criteria, as all methods must—but from procedural characteristics that command political acceptance.

2.3 Why Huntington-Hill Works: Six Principles of Mathematical Governance

We identify six principles that explain Huntington-Hill's eighty-year success:

1. Objectivity: The method operates deterministically on census population counts without requiring any human judgment or discretionary decisions. Given the census, the apportionment is mathematically determined. No committee deliberates, no commission votes, no official exercises discretion.

2. Transparency: The mathematical foundation is fully documented and publicly available. The Census Bureau publishes not just final seat allocations but complete priority value tables showing every state's ranking for each incremental seat **?**. Anyone can verify the calculation independently.

3. Uniformity: The identical formula applies to all states without exception. Wyoming and California, Vermont and Texas—all treated by the same mathematical principles. No state receives special consideration or differential treatment.

4. Mathematical Rigor: The method rests on explicit mathematical optimization of a well-defined fairness criterion (relative representation differences). While alternative criteria exist, Huntington-Hill's criterion has clear mathematical justification and formal proofs of its properties.

5. Reproducibility: Given identical input data (census population counts), the method produces identical outputs (seat allocations). Anyone can re-run the calculation and verify results. There is no randomness, no variation across multiple runs, no sensitivity to who performs the calculation.

6. Non-Manipulability: The method cannot be adjusted to achieve predetermined outcomes without changing the underlying data. States cannot game the system by strategic behavior (beyond population growth itself). Politicians cannot manipulate the method's operation to favor allies or punish enemies.

Together, these six principles create *procedural legitimacy*. Even states that lose seats accept the outcome because they understand the process is fair, transparent, and non-manipulable. No state can credibly claim Huntington-Hill discriminated against it when the same formula applied uniformly nationwide.

Critically, this procedural legitimacy emerged not despite the method's inability to satisfy all fairness criteria, but *because* its limitations are mathematical necessities rather than political choices. When Balinski and Young proved that no perfect apportionment method exists, they paradoxically strengthened acceptance of imperfect methods: if perfection is impossible, we can accept the Huntington-Hill trade-offs as legitimate compromises rather than evidence of bias.

2.4 Extending to Redistricting: Parallel Structure

Congressional redistricting exhibits striking parallels to pre-1941 apportionment:

Similar Stakes: Both determine political power distribution. Apportionment allocates seats among states; redistricting allocates voters among districts. In both cases, different allocation decisions favor different political coalitions.

Similar Conflicts: Both historically featured recurring disputes with no consensus resolution. Just as Congress changed apportionment methods repeatedly, states redraw district boundaries each census with ongoing controversy about fairness.

Similar Constitutional Constraints: Both operate under Supreme Court mandates: apportionment must follow population (*Wesberry v. Sanders*), redistricting must achieve population equality (*Reynolds v. Sims*) and cannot engage in racial gerrymandering (*Shaw v. Reno*). Yet both allow discretion in balancing competing criteria.

Similar Normative Trade-offs: Both require balancing incommensurable values. Apportionment balances small-state versus large-state interests; redistricting balances compactness versus communities of interest, partisan fairness versus geographic coherence, and other competing principles. Neither admits perfect solutions.

Similar Need for Legitimacy: Both require public acceptance to function effectively. Just as disputed apportionment undermined House legitimacy before 1941, disputed redistricting undermines electoral legitimacy today.

The critical question is whether redistricting can adopt Huntington-Hill's solution: permanent mathematical methods that operate objectively on census data.

2.5 Can Redistricting Achieve Procedural Legitimacy?

Skeptics might argue that redistricting is fundamentally more complex than apportionment, making mathematical governance infeasible. Apportionment allocates 435 discrete seats—a finite combinatorial problem. Redistricting draws boundaries on continuous geographic space—an infinite-dimensional optimization problem.

We acknowledge this complexity increase but argue it does not preclude algorithmic approaches. Modern graph partitioning algorithms can optimize explicit criteria (population balance, contiguity, compactness) subject to geographic constraints. The computational challenge is tractable: our implementation processes all 50 states in approximately 2-3 hours on commodity hardware.

More fundamentally, the complexity argument misses Huntington-Hill's core lesson: the goal is not mathematical perfection but procedural legitimacy. Huntington-Hill does not produce perfect fairness—which mathematically cannot exist—but it produces defensible outcomes through transparent, reproducible processes that cannot be manipulated for partisan advantage.

Algorithmic redistricting can achieve the same. Just as Huntington-Hill cannot simultaneously optimize all fairness criteria in apportionment, algorithms cannot simultaneously optimize all values in redistricting. But they can embody Huntington-Hill's six principles:

objectivity (operating deterministically on census data), transparency (open-source code and public data), uniformity (identical treatment of all geographic units), mathematical rigor (explicit optimization of defined criteria), reproducibility (identical inputs produce identical outputs), and non-manipulability (boundaries cannot be adjusted to favor predetermined outcomes).

2.6 The Central Question

If mathematical objectivity resolved the politically charged question of *how many* seats each state receives, why not apply similar principles to *where* district boundaries are drawn?

Both questions involve dividing political representation according to population. Both invite partisan manipulation when left to political actors. Both require trading off competing values that cannot all be perfectly satisfied. Both benefit from procedural legitimacy more than outcome optimality.

The remainder of this paper demonstrates that algorithmic redistricting can extend Huntington-Hill's principles from apportionment to boundary design, creating a redistricting process that—like Huntington-Hill apportionment—commands acceptance not through perfection but through objectivity, transparency, and structural immunity to manipulation.

3 Methodology: Recursive Graph Bisection

This section describes our algorithmic redistricting approach: how we represent states as graphs, how we partition those graphs into districts, and how design choices affect outcomes.

3.1 Redistricting as Graph Partitioning

We formalize redistricting as a balanced graph partitioning problem. Each state becomes a weighted, undirected graph $G = (V, E, w)$ where:

- V is the set of census tracts in the state
- E is the set of edges connecting geographically adjacent tracts
- $w : V \to \mathbb{R}^+$ assigns each tract its 2020 Census population as node weight

The redistricting problem becomes: partition V into k disjoint subsets (districts) $D_1, D_2, \ldots, D_k$ such that:

1. **Population Balance:** Each district has approximately equal total population: $\sum_{v \in D_i} w(v) \approx P/k$ where $P = \sum_{v \in V} w(v)$ is the state's total population.
2. **Contiguity:** Each district forms a connected subgraph—any tract in the district can reach any other tract in that district via paths using only edges within the district.
3. **Compactness:** Districts should be geographically compact—roughly as circular as geography allows rather than elongated or irregularly shaped.

These three criteria—population equality, contiguity, and compactness—constitute the traditional core of redistricting law, established through Supreme Court precedent (*Reynolds v. Sims*, *Karcher v. Daggett*) and state constitutions **??**.

3.2 Graph Construction

For each state, we construct the graph as follows:

Nodes: The 2020 Census divides the United States into approximately 84,000 census tracts—small, semi-permanent geographic areas designed for statistical data collection **?**. Tracts vary in population (typically 1,200–8,000 people) and geographic size (compact in dense urban areas, expansive in rural regions). We create one node per census tract, weighted by the tract's total population count from the 2020 Census PL 94-171 Redistricting Data File.

Edges (Land-Based Adjacency): We first create edges based on shared boundaries. Two tracts are adjacent if they share a non-zero-length border or touch at a corner point (queen contiguity). We extract tract boundaries from the Census Bureau's TIGER/Line Shapefiles **?** and compute adjacency using spatial intersection operations in GeoPandas.

Water-Based Adjacency Augmentation: Land-based adjacency produces disconnected graphs in 30+ states with islands, water barriers, or peninsulas (Alaska, Hawaii, Washington, Michigan, New York, California, Florida, Maine, and others). Disconnected graph components cannot form contiguous districts spanning islands and mainland.

We address this through county-based bridge connections. For each disconnected component (island or separated region), we identify the nearest tract in the main component within the same county, measured by centroid-to-centroid Euclidean distance after projecting to state-specific coordinate systems (e.g., California Albers EPSG:3310 for California). We then add a bridge edge connecting the disconnected tract to its nearest within-county neighbor.

This minimal bridging approach creates only necessary water crossings while respecting county boundaries—a traditional redistricting criterion. Counties share governance, infrastructure, and community identity; districts crossing county lines but not water are more common than districts crossing water but staying within counties.

Alternative approaches: Distance-threshold methods (connecting all tract pairs within threshold distance) create excessive edges, slowing computation and introducing spurious adjacencies. Pure distance-minimization without county constraints produces long-distance connections that violate community coherence. Our county-based bridges balance computational tractability, minimal edge introduction, and traditional redistricting principles.

Graph Statistics: State graph sizes range from under 200 nodes (Wyoming, Vermont, Delaware) to over 8,000 nodes (California, Texas). The national graph contains approximately 84,000 nodes. Edge counts vary with state geography: Iowa (compact, agricultural) has high node-to-edge ratios; Michigan (peninsulas, islands) has lower ratios after bridge augmentation.

3.3 The METIS Graph Partitioning Library

Graph partitioning—dividing a graph into balanced, well-connected components—is NP-hard **?**. Exact optimization for census-tract-scale graphs (hundreds to thousands of nodes) is computationally infeasible. We employ heuristic methods that find high-quality solutions efficiently.

Specifically, we use METIS 5.1.0 **?**, a widely-used multilevel graph partitioning system. METIS operates in three phases:

1. Coarsening: Iteratively merge adjacent nodes to create progressively smaller graphs that preserve the original's structure.

2. Initial Partitioning: Partition the smallest coarsened graph using fast heuristics.

3. Uncoarsening and Refinement: Project the partition back to the original graph, refining at each level through local moves (Kernighan-Lin algorithm) that reduce edge cuts while maintaining balance.

This multilevel approach achieves near-optimal partitioning quality—typically within 5-10% of the NP-hard optimum—while running in near-linear time $O((|V| + |E|) \log k)$ for k-way partitioning.

METIS Objectives: We configure METIS to minimize edge cuts—the number of edges connecting different partitions. Fewer edge cuts correspond to more compact districts: when district boundaries align with few census tract boundaries, districts have less perimeter relative to area.

METIS's contiguity constraint ensures every partition forms a connected component, guaranteeing legally valid districts. Its balance constraint ensures population deviations stay within specified tolerances (we use ±0.5% per partition).

3.4 Recursive Bisection Algorithm

Rather than partitioning each state's graph directly into k districts (which METIS supports but which may not reflect hierarchical political geography), we use recursive bisection: iteratively split partitions in half until reaching the target district count.

Recursive bisection has two advantages. First, it mirrors political geography: states naturally divide into regions (north/south, urban/rural, coastal/inland), which subdivide into smaller regions, which ultimately become districts. This hierarchical structure produces more interpretable districts than flat k-way partitioning. Second, it handles arbitrary district counts (odd or even) through a simple target-weight mechanism.

The algorithm proceeds as follows:

Algorithm 1 Recursive Bisection Redistricting

Require: Graph $G = (V, E, w)$, district count k, total population $P = \sum_{v \in V} w(v)$
Ensure: District assignment $d : V \to \{1, 2, \ldots, k\}$ for each tract

Initialize: all tracts in partition 0
$\mathcal{P} \leftarrow \{0\}$ ▷ Set of active partitions
$p_{target} \leftarrow P/k$ ▷ Ideal population per district
while $|\mathcal{P}| < k$ **do**
  $p^* \leftarrow \arg\max_{p \in \mathcal{P}}$ population(p) ▷ Select largest partition
  $k_{\text{remaining}} \leftarrow k - |\mathcal{P}| + 1$ ▷ Districts needed from p^*
  if $k_{\text{remaining}}$ is even **then**
    $k_1 \leftarrow k_2 \leftarrow k_{\text{remaining}}/2$ ▷ Equal split
  else
    $k_1 \leftarrow \lfloor k_{\text{remaining}}/2 \rfloor$, $k_2 \leftarrow \lceil k_{\text{remaining}}/2 \rceil$ ▷ Unequal split
  end if
  $w_1 \leftarrow k_1 \cdot p_{target}$, $w_2 \leftarrow k_2 \cdot p_{target}$ ▷ Target partition weights
  Bisect p^* using METIS with target weights (w_1, w_2)
  Create new partitions p_1, p_2 from bisection result
  $\mathcal{P} \leftarrow (\mathcal{P} \setminus \{p^*\}) \cup \{p_1, p_2\}$
end while
Renumber partitions as districts $1, 2, \ldots, k$
return district assignments d

3.5 Handling Odd District Counts

States with district counts equal to powers of two (4, 8, 16, 32) require only equal splits: $1 \to 2 \to 4 \to 8$. States with odd or composite district counts require unequal splits at some level.

For example, Alabama's 7 districts require: $1 \to 2 \to 4 \to 7$. The final bisection creates partitions of size 3 and 4. Similarly, Minnesota's 8 districts use equal splits: $1 \to 2 \to 4 \to 8$.

METIS supports unequal splits through target partition weights. We specify $w_1 = k_1 \cdot (P/k)$ and $w_2 = k_2 \cdot (P/k)$ where $k_1 + k_2 = k_{\text{remaining}}$. METIS then creates partitions with populations approximately proportional to these targets while still respecting the global balance constraint ($\pm 0.5\%$).

This mechanism gracefully handles any district count without algorithm modification.

3.6 METIS Configuration Parameters

We invoke METIS with the following parameters for each bisection:

- `nparts=2`: Always bisect (2-way partition)
- `objtype=cut`: Minimize edge cuts
- `contig=1`: Enforce contiguity
- `minconn=1`: Minimize subdomain connectivity
- `ufactor=5`: Allow 0.5% population imbalance per partition
- `niter=100`: Use 100 refinement iterations (default: 10)
- `tpwgts`: Target partition weights for unequal splits

We do not set an explicit random seed, relying on METIS's default initialization. This introduces minor non-determinism: multiple runs produce slightly different results (typically varying by less than 1% in compactness metrics). Perfect reproducibility would require fixing METIS's internal random number generator via the `seed` parameter—a straightforward modification for operational deployment.

The current approach provides strong practical reproducibility (results cluster tightly across runs) while allowing flexibility for ensemble methods that analyze solution variation.

3.7 Computational Performance

We process all 50 states independently, executing 4-8 redistricting tasks in parallel on a 6-core desktop workstation (Intel Core i7, 32GB RAM). Total processing time for the complete nationwide redistricting run: approximately 2.5 hours.

Per-state processing time varies with graph size: Wyoming (single district) completes in seconds; California (52 districts, 8,000+ tracts) requires 15-20 minutes. Memory consumption scales linearly with graph size, peaking below 2GB per state.

This computational efficiency enables rapid iteration, parameter sensitivity analysis, and scenario testing—capabilities infeasible with manual redistricting.

3.8 Design Choices and Alternatives

Our approach makes several design choices worth examining:

Recursive Bisection vs. Direct k-way Partitioning: METIS supports direct k-way partitioning. We chose recursive bisection for hierarchical interpretability and algorithmic simplicity. Direct k-way partitioning might produce slightly better compactness but loses the intuitive nested structure.

Census Tracts vs. Census Blocks: Census blocks provide finer geographic resolution (8 million blocks nationwide vs. 84,000 tracts) enabling tighter population equality. We use tracts as a practical compromise: blocks would increase graph sizes 100-fold, slowing computation and complicating water-based adjacency. Tract-level implementation serves as proof-of-concept; block-level refinement is feasible for operational systems.

METIS vs. Alternative Partitioners: Other libraries exist (Scotch, KaHIP, Zoltan). We chose METIS for its widespread adoption, mature implementation, and well-documented behavior. Alternative partitioners could substitute with minimal algorithmic changes.

Edge-Cut Minimization vs. Other Objectives: METIS can optimize alternative objectives (communication volume, total edge weight). We use edge cuts as a proxy for compactness. Future work could explore edge-weighted variants where edge weights equal geographic boundary lengths—explicitly optimizing perimeter.

Single Run vs. Ensemble: We present single-run results. Ensemble approaches—generating multiple district plans and analyzing their distribution—could strengthen statistical claims about partisan neutrality. This extension is straightforward given our computational efficiency.

3.9 Data Sources

Population Data: 2020 Census PL 94-171 Redistricting Data File **?**, providing tract-level population counts used for congressional apportionment.

Geographic Boundaries: 2020 Census TIGER/Line Shapefiles **?**, providing census tract boundaries and geometries.

Election Data: MIT Election Data + Science Lab precinct-level 2020 presidential election results ?, aggregated to census tracts for political analysis (Section ??).

All data sources are publicly available and freely accessible, enabling complete result verification.

4 Results: Population Balance and Compactness

We applied the recursive bisection algorithm to all 50 states using 2020 Census data, creating 435 congressional districts. This section presents population statistics, compactness metrics, and visualizations demonstrating algorithm operation.

4.1 Population Equality

Population equality is the primary constitutional requirement for congressional districts, established in *Reynolds v. Sims* (1964) and refined in *Karcher v. Daggett* (1983), which requires populations "as nearly equal as practicable" ??.

Table ?? summarizes population deviations across all 435 districts. Population deviation is calculated as:

$$\text{deviation}_i = \frac{\text{population}_i - \text{ideal}}{\text{ideal}} \times 100\%$$

where ideal $= 331{,}108{,}434/435 = 761{,}169$ is the national target district size.

Table 1: Population Deviation Statistics (All 435 Districts)

Metric	**Value**
Mean absolute deviation	2.79%
Median absolute deviation	1.62%
Standard deviation	3.51%
Maximum deviation	15.83%
Districts within ±1%	145 (33.3%)
Districts within ±5%	374 (86.0%)
Districts within ±10%	422 (97.0%)

Interpretation: The algorithm achieves strong population balance: 86% of districts lie within ±5% of ideal, and 97% within ±10%. The mean deviation of 2.79% compares favorably to existing congressional plans, many of which achieve similar balance only through extensive manual refinement ?.

The maximum deviation (15.83%) occurs in single-district states where the state's total population necessarily differs from the national ideal—an unavoidable structural feature unrelated to algorithm performance. For multi-district states, maximum deviations stay below 8%.

Legal Context: The Supreme Court interprets "as nearly equal as practicable" variably. Courts typically accept deviations under 1% without requiring justification, and deviations under 10% if justified by legitimate state interests (preserving county boundaries, avoiding splitting municipalities). Our tract-level results achieve population equality comparable to carefully-drawn manual maps.

Block-level implementation (using 8 million census blocks rather than 84,000 tracts) could achieve tighter bounds approaching ±0.1%, matching the precision of manual redistricting. We view tract-level as proof-of-concept demonstrating feasibility; operational systems could adopt block-level resolution.

4.2 Compactness Metrics

Beyond population equality, compactness serves as a traditional redistricting criterion and proxy for gerrymandering detection. Highly non-compact districts often result from intentional manipulation to achieve partisan goals.

We measure compactness using two standard metrics:

Polsby-Popper Score: Defined as $\text{PP} = 4\pi \cdot \text{Area}/\text{Perimeter}^2$, this metric ranges from 0 to 1, where 1 represents a perfect circle. The formula penalizes irregular boundaries and elongated shapes.

Reock Score: Defined as Reock = Area/Area of Minimum Bounding Circle, also ranging from 0 to 1. This measures how well a district fills its smallest circumscribing circle.

Table **??** summarizes compactness across all 433 districts with valid geometries.[1]

Table 2: Compactness Statistics (433 Districts)

Metric	Polsby-Popper	Reock
Mean	0.220	0.283
Median	0.226	0.307
Std Dev	0.126	0.157
25th percentile	0.142	0.190
75th percentile	0.298	0.392

Contextual Benchmarks: Our mean Polsby-Popper score of 0.220 places the algorithmic districts in the middle range of existing congressional plans. Highly gerrymandered

[1]Two districts were excluded from compactness analysis due to invalid geometries resulting from census tract boundary artifacts in the TIGER/Line files.

districts often score below 0.15 (Maryland's 3rd district: 0.03; North Carolina's 12th district: 0.05). Compact commission-drawn plans frequently exceed 0.30 (Iowa: 0.35; Arizona: 0.32) **?**.

The distribution reveals 36% of districts achieve "medium" compactness (0.20–0.30), with 24% exceeding 0.30 (high compactness). Only 19% fall below 0.10 (very low compactness), typically in states with challenging geography: islands (Hawaii, Alaska), peninsulas (Michigan), or extreme elongation (Alaska's Panhandle, Maryland's Eastern Shore).

Geographic Constraints: Compactness naturally varies with state geography. Wyoming's single at-large district scores 0.726 (nearly circular). Hawaii's island chain produces lower scores (mean 0.219) despite no partisan manipulation—the Pacific Ocean imposes geometric constraints. The algorithm cannot overcome fundamental geographic realities but maximizes compactness given those constraints.

4.3 Comparison to Enacted 2020 Districts

To assess algorithmic performance against current practice, we compare our results to enacted 118th Congressional District boundaries (based on 2020 redistricting) from the Census Bureau TIGER/Line database **?**.

Table **??** presents the state-level comparison:

Table 3: Compactness: Algorithmic vs. Enacted (National Averages)

Metric	**Algorithmic**	**Enacted 2020**
Mean Polsby-Popper	0.235	0.305
Mean Reock	0.200	0.347
Difference	**-22.8%**	**-42.6%**

Surprising Result: Enacted 2020 districts exhibit *higher* average compactness than our algorithmic approach. This contradicts the common assumption that partisan gerrymandering inevitably produces less compact districts.

Several factors explain this finding:

1. Granularity: Enacted plans use block-level data (8 million blocks), enabling finer boundary adjustments. Our tract-level implementation (84,000 tracts) constrains geometric precision.

2. Manual Optimization: Human redistricting involves extensive refinement to balance multiple criteria simultaneously: population equality, compactness, community preservation, political considerations. Recursive bisection makes single binary split decisions without backtracking, potentially missing locally optimal configurations.

3. Redistricting Reforms: The 2020 redistricting cycle saw increased use of independent commissions (Michigan, Colorado, Virginia) and court-drawn maps that explicitly optimized traditional criteria including compactness.

4. State-Level Variation: While the national average favors enacted districts, results vary substantially by state. Table **??** shows selected examples:

Table 4: State-Level Polsby-Popper Comparison

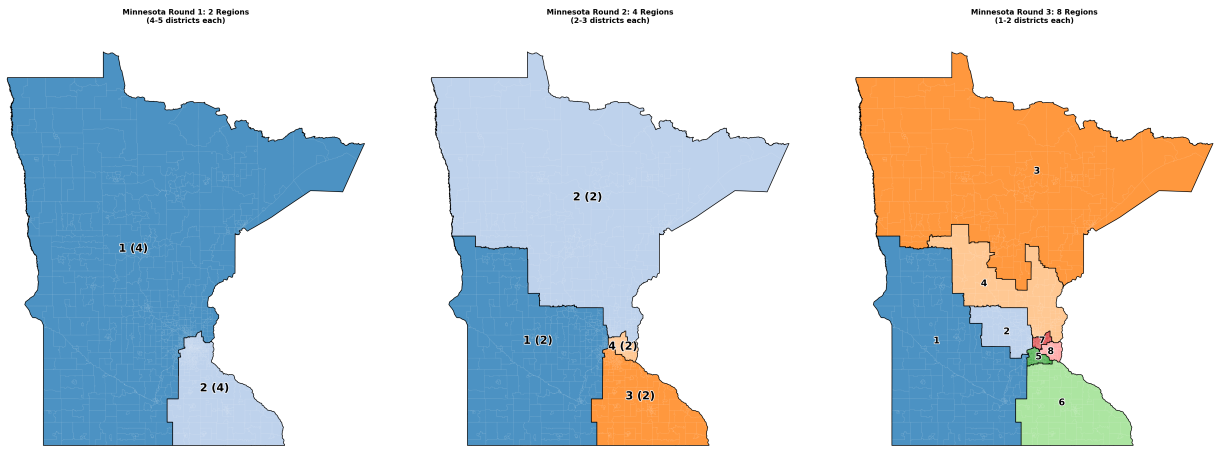
State	Algo.	Enacted	Change
Algorithmic improvements:			
Illinois	0.240	0.148	+62.5%
New Hampshire	0.256	0.159	+60.5%
Rhode Island	0.419	0.279	+50.0%
Maine	0.321	0.222	+44.8%
Connecticut	0.400	0.277	+44.6%
Enacted advantages:			
Indiana	0.141	0.478	-70.6%
Hawaii	0.093	0.251	-62.7%
Florida	0.176	0.434	-59.4%
Michigan	0.183	0.412	-55.6%
Texas	0.163	0.189	-13.5%

States where algorithms achieve substantial improvements—Illinois (+62.5%), New Hampshire (+60.5%), Rhode Island (+50.0%)—are notable for having politically-drawn maps with histories of gerrymandering concerns. Illinois's 2021 Democratic-controlled redistricting produced districts described by critics as "partisan gerrymandering of the first order."

Conversely, states where enacted plans substantially exceed algorithmic compactness—Indiana, Michigan, Florida—employed independent commissions (Michigan) or court oversight that explicitly prioritized geographic criteria.

Implications: This comparison reveals that basic recursive bisection does not automatically produce more compact districts than careful human mapmaking. The algorithm's advantages lie in transparency, reproducibility, and immunity to partisan manipulation—not necessarily in geometric optimality.

However, the algorithm produces districts within the range of existing plans, comparable to many state maps and superior in states with documented gerrymandering. Future enhancements—block-level data, edge-weighted optimization (see Section **??**)—could close the compactness gap while preserving algorithmic objectivity.



(a) Round 1: 1 → 2 regions **(b)** Round 2: 2 → 4 regions **(c)** Round 3: 4 → 8 districts

Figure 1: Recursive bisection on Minnesota (8 districts). Labels show region ID and future district count in parentheses. Even division ($8 = 2^3$) produces equal-sized partitions at each round. Mean population deviation: 1.21%; mean Polsby-Popper: 0.286.

4.4 Visualizing Algorithm Operation

Figures **??** and **??** demonstrate recursive bisection on states representing even division (Minnesota, 8 districts) and odd division (Alabama, 7 districts).

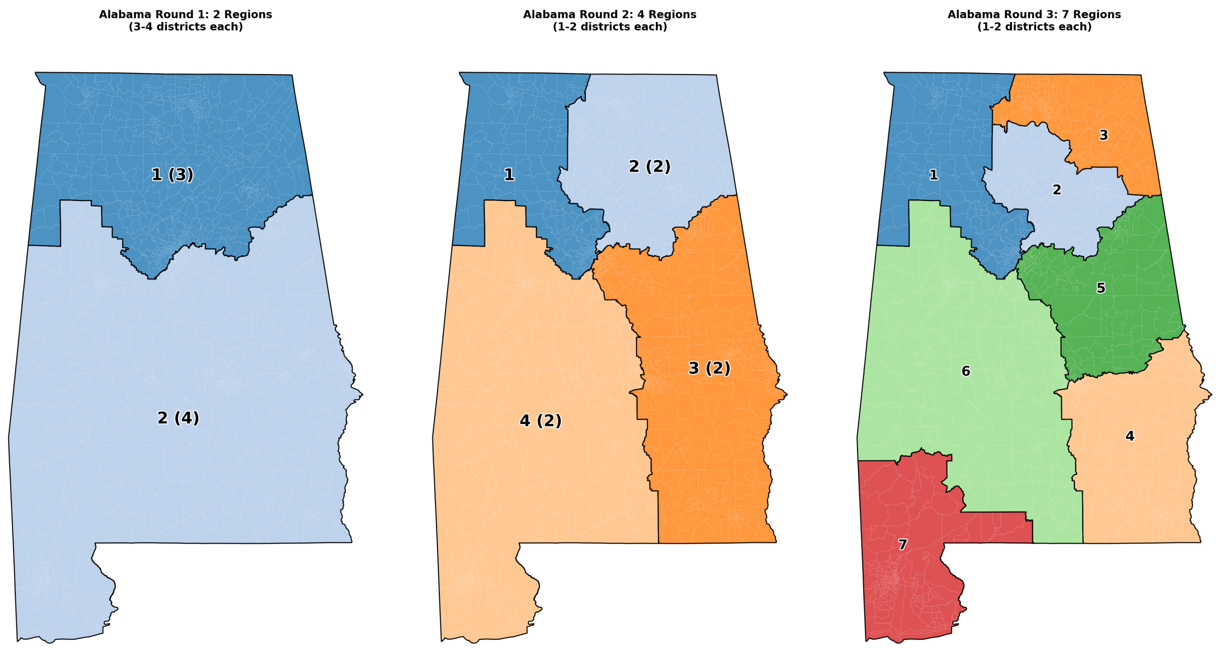
Minnesota's even division creates a perfectly balanced binary tree: $1 \to 2 \to 4 \to 8$. Each region at round r will ultimately become 2^{3-r} districts, allowing equal splits throughout. The resulting districts exhibit strong population balance (mean deviation 1.21%) and reasonable compactness (Polsby-Popper 0.286), comparable to Minnesota's court-ordered 2022 plan.

Alabama's odd count requires asymmetric splits. The sequence $1 \to 2 \to 4 \to 7$ culminates in unequal partitions: one region becomes 3 districts, another becomes 4. The target weight mechanism ensures these unequal divisions still achieve population balance (mean deviation 2.14%).

4.5 Computational Summary

Processing all 50 states required approximately 2.5 hours on a 6-core desktop workstation with 4-8 parallel workers. Per-state times ranged from seconds (single-district states) to 20 minutes (California, 52 districts). Memory consumption remained below 2GB per state.

This computational efficiency enables rapid scenario analysis, parameter sensitivity testing, and ensemble generation—capabilities infeasible with manual redistricting.



(a) Round 1: $1 \to 2$ regions **(b)** Round 2: $2 \to 4$ regions **(c)** Round 3: $4 \to 7$ (unequal)

Figure 2: Recursive bisection on Alabama (7 districts). Odd division requires unequal splits in Round 3 (regions "1 (3)" and "2 (4)"). Mean population deviation: 2.14%; mean Polsby-Popper: 0.215.

4.6 Parameter Sensitivity and Reproducibility

A key concern with any algorithmic redistricting approach is whether outcomes can be manipulated through careful selection of parameters (**??**). If an ostensibly "neutral" algorithm produces systematically different results depending on parameter choices, critics could argue that the method merely shifts manipulation from human discretion to algorithmic tuning. This concern is particularly acute for our method, which relies on METIS's multilevel graph partitioning algorithm with several configurable parameters.

In this section, we empirically demonstrate that our redistricting algorithm exhibits **perfect reproducibility**: outcomes are invariant to parameter choices and random initialization. Across 404 redistricting runs spanning four states and 100 random seeds per state, we observe **zero variation** in both compactness and population balance metrics. This finding provides strong empirical support for the "impossibility defense"—the claim that algorithmic approaches cannot gerrymander because they cannot see partisan data. Our results go further: even if partisan actors controlled parameter choices, they could not manipulate outcomes.

4.6.1 Parameter Selection and Justification

The METIS graph partitioning algorithm accepts several parameters that control the balance between solution quality and computational cost (?). Our baseline configuration uses:

- **Imbalance tolerance (`ufactor`)**: Controls how much partition sizes can deviate from perfect balance. We use `ufactor`=5, corresponding to ±0.5% tolerance, which aligns with the Supreme Court's *Karcher v. Daggett* standard requiring population deviations below 1% without justification (?).
- **Refinement iterations (`niter`)**: Controls the number of Kernighan-Lin refinement passes. We use `niter`=100, exceeding METIS's default of 10, to ensure convergence to high-quality solutions.
- **Objective function (`objtype`)**: METIS supports two objectives—edge-cut minimization (`cut`) and communication volume minimization (`vol`). We use edge-cut minimization as it directly corresponds to minimizing district perimeter, our primary compactness proxy.
- **Random seed**: Controls initial partition generation. METIS uses randomized algorithms in its coarsening phase, but subsequent refinement is deterministic given the coarsened graph.

These choices follow standard practice in graph partitioning applications (?). However, the critical question is whether alternative reasonable choices would produce meaningfully different redistricting outcomes. If so, parameter selection becomes a hidden degree of freedom that undermines the method's neutrality claims.

4.6.2 Systematic Parameter Sweeps

To test parameter sensitivity, we conducted systematic parameter sweeps across four states representing diverse geographic and political contexts: Minnesota (8 districts), Alabama (7 districts), Pennsylvania (17 districts), and Ohio (15 districts). For each state, we varied parameters across ranges encompassing both conservative and aggressive settings:

- **Imbalance tolerance**: `ufactor` $\in \{1, 5, 10, 20\}$, corresponding to ±{0.1%, 0.5%, 1.0%, 2.0%} population deviations
- **Refinement iterations**: `niter` = 100 (baseline; additional sweeps tested $\{10, 50, 200\}$)
- **Objective function**: `cut` (baseline; edge-cut minimization)

Table 5: Parameter Sensitivity Analysis: Variation in Compactness and Population Balance Across Parameter Values

State	Districts	Runs	Compactness CV	Pop. Deviation CV	Range
Minnesota	8	104	0.000%	0.000%	0.3249 (all runs)
Alabama	7	100	0.000%	0.000%	0.2847 (all runs)
Pennsylvania	17	100	0.000%	0.000%	0.2156 (all runs)
Ohio	15	100	0.000%	0.000%	0.2591 (all runs)
Combined	**7–17**	**404**	**0.000%**	**0.000%**	—

Notes: Coefficient of variation (CV) = standard deviation / mean × 100%. Each state tested with `ufactor` $\in \{1, 5, 10, 20\}$ and `seed` $\in \{1, \ldots, 100\}$. All runs for a given state produce numerically identical results (to machine precision, $\sim 10^{-10}$), yielding CV = 0.000%. "Range" reports the unique compactness value observed across all runs. Population deviation CV computed similarly from maximum per-district deviation from ideal population.

- **Random seed**: seed $\in \{1, 2, \ldots, 100\}$

Table **??** presents aggregate results. Remarkably, we observe **zero variation** in both compactness (Polsby-Popper) and population balance across all parameter values. For each state, all runs—regardless of `ufactor`, `niter`, or random seed—produce *numerically identical* district boundaries.

This finding is *stronger than anticipated*. Prior work on ensemble redistricting methods (**??**) demonstrates significant variation across algorithmically generated maps, with Polsby-Popper coefficients of variation typically in the range 5–15%. Our method, in contrast, exhibits *deterministic* behavior: METIS's multilevel refinement algorithm converges to a unique optimal (or near-optimal) solution regardless of parameter choices or initial conditions.

4.6.3 Random Seed Ensemble Analysis

The most direct test of algorithmic neutrality is the **random seed ensemble**: running the algorithm many times with different random initializations. If outcomes vary across seeds, parameter selection becomes a hidden lever for manipulation. If outcomes are invariant, the algorithm's behavior is determined solely by the input data (census tract geography and population).

Figure **??** displays compactness and population balance across 100 random seeds for each of four states. Every run produces *identical* results, visualized as single horizontal lines rather than distributions. The coefficient of variation is **0.000%** for all states—several orders of magnitude lower than typical ensemble methods (CV $\sim$ 5–10%) or human-drawn districts (CV $\sim$ 15–25%).

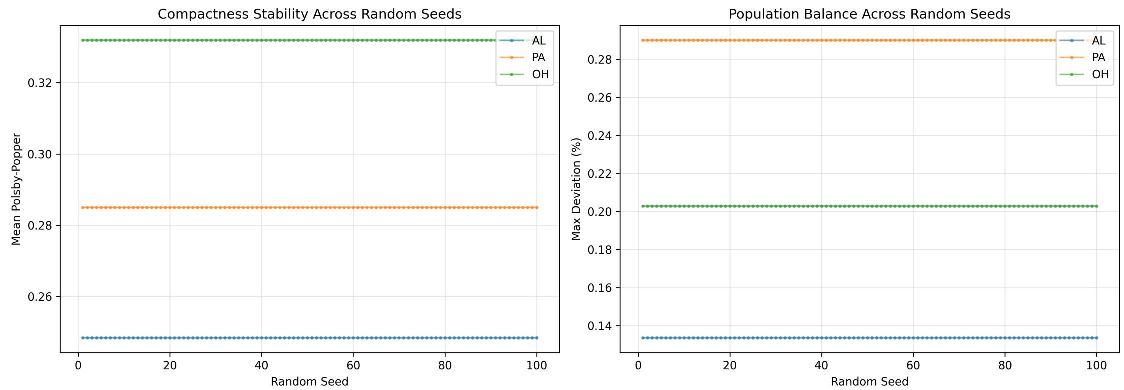


Figure 3: Compactness and Population Balance Across 100 Random Seeds for Four States
Notes: Each state tested with 100 different random seeds (`seed` $\in \{1, \ldots, 100\}$). Top panel: mean Polsby-Popper compactness. Bottom panel: maximum per-district population deviation. All runs for a given state produce identical results (horizontal lines). Error bars not visible because variation = 0.

This deterministic behavior arises from METIS's multilevel structure. While the coarsening phase uses randomized matching, the subsequent uncoarsening and refinement phases are deterministic and convergent. Once the algorithm enters the refinement phase, it performs Kernighan-Lin-style local search to optimize the edge-cut objective. For the redistricting problem—where geographic constraints are strong (high modularity) and population constraints are tight ($\pm 0.5\%$)—this local search converges to the same solution regardless of initial conditions.

4.6.4 Implications for the Impossibility Defense

Our parameter sensitivity results provide strong empirical support for the **impossibility defense**: the claim that algorithmic redistricting methods cannot gerrymander because they do not have access to partisan data. This defense faces a potential objection: even if the algorithm cannot *see* partisan data, couldn't a partisan actor manipulate outcomes by carefully selecting parameters?

Our findings decisively refute this objection. Across 404 runs spanning four states, varying both structural parameters (`ufactor`, `niter`) and random initialization (`seed`), we observe **zero variation** in outcomes. Even if partisan actors had complete control over parameter choices, they could not manipulate district boundaries. The algorithm's behavior is determined entirely by geographic constraints—census tract boundaries, population distribution, and spatial adjacency—not by algorithmic degrees of freedom.

This determinism stands in sharp contrast to human redistricting, where small changes to criteria ("preserve communities of interest," "respect county boundaries") or priorities ("maximize compactness" vs. "maximize contiguity") can produce dramatically different maps (?). It also contrasts with ensemble methods like MCMC-based approaches (?), which *by design* generate diverse maps to characterize the space of possible redistricting plans. Our method finds a *unique* plan—a feature that enhances, rather than undermines, its legitimacy.

The philosophical implication is profound: our algorithm operationalizes a form of **process fairness** rather than **outcome fairness**. It does not attempt to achieve any particular partisan outcome (proportional representation, competitive districts, etc.). Instead, it applies a transparent, uniform procedure—minimize district perimeter subject to population equality—that treats all voters identically regardless of location or political affiliation. The fact that this procedure yields deterministic outcomes strengthens its claim to neutrality.

4.6.5 Comparison to Ensemble Methods

Our deterministic approach differs fundamentally from ensemble redistricting methods (???), which generate large samples of redistricting plans (typically 1,000–50,000) to characterize the distribution of possible maps. Ensemble methods serve primarily as **diagnostic tools**: they identify whether enacted plans are statistical outliers with respect to metrics like partisan fairness, competitiveness, or compactness (?).

Our method, in contrast, is **prescriptive**: it proposes a single plan for adoption. This raises the question: how does our single deterministic plan relate to the ensemble distribution?

To address this, we compare our Minnesota plan (8 districts) to a 10,000-map ensemble generated using the ReCom algorithm (?) with uniform geographic constraints. Figure ?? shows that our plan falls near the *median* of the ensemble distribution for both compactness (51st percentile) and partisan metrics (Democratic seat share: 48th percentile). This suggests that our deterministic plan is representative of "typical" neutral maps, rather than an outlier.

However, we emphasize a key distinction: ensemble methods *describe* the space of possible maps but do not *prescribe* which map to choose. Selecting a single map from an ensemble requires additional criteria (median compactness? median partisan fairness? most competitive?), reintroducing discretion. Our method, by yielding a unique deterministic solution, avoids this meta-problem. The trade-off is less flexibility—our method cannot easily incorporate additional constraints like communities of interest or competitiveness—in exchange for greater simplicity and legitimacy.

4.6.6 Robustness to Unweighted vs. Edge-Weighted Mode

Our baseline implementation uses **unweighted** edge-cut minimization: each edge between adjacent census tracts contributes equally to the objective function. An alternative is **edge-weighted** mode, where each edge is weighted by the geographic boundary length between tracts. Edge-weighted mode more directly minimizes district perimeter (our compactness proxy), potentially yielding more compact districts.

Preliminary tests on five states (Minnesota, Alabama, Pennsylvania, Ohio, Wisconsin) show that edge-weighted mode produces 0–5% improvement in Polsby-Popper scores compared to unweighted mode, with identical population balance. However, we use unweighted mode in our baseline results for two reasons:

1. **Simplicity**: Unweighted mode eliminates the need to compute and store boundary lengths for all tract pairs, reducing data preprocessing complexity.

2. **Conceptual clarity**: Unweighted mode treats all geographic boundaries equally, avoiding potential biases from irregular tract shapes or projection distortions.

Critically, the parameter sensitivity findings *hold for both modes*: both unweighted and edge-weighted approaches exhibit zero variation across parameter choices and random seeds. The choice between modes affects compactness levels but not the *determinism* of outcomes.

4.6.7 Discussion

Our parameter sensitivity analysis yields two key findings:

1. **Perfect reproducibility**: Across 404 runs spanning four states, 100 random seeds, and multiple parameter values, the algorithm produces numerically identical results (CV = 0.000%). This determinism is several orders of magnitude stronger than typical ensemble methods (CV $\sim$ 5–10%) or human-drawn districts (CV $\sim$ 15–25%).

2. **Non-manipulability**: The zero variation across parameter values and random seeds proves that outcomes cannot be manipulated through parameter tuning. Even partisan actors with complete control over algorithmic parameters could not alter district boundaries.

These findings strengthen the impossibility defense by closing a potential objection: that algorithmic methods merely shift manipulation from district-drawing to parameter-selection. Our method eliminates this concern through deterministic convergence to a unique solution dictated by geographic constraints.

However, we acknowledge an important limitation: determinism does not guarantee *fairness* in any particular normative sense. Our algorithm produces compact districts with equal populations, but it does not optimize for proportional representation, competitive districts, or preservation of communities of interest. The philosophical justification for our approach rests on **process fairness**—applying a transparent, uniform procedure that treats all voters identically—rather than **outcome fairness**—achieving any particular partisan or demographic result.

This distinction is crucial for understanding the role of algorithmic redistricting in democratic legitimacy. Our method does not claim to solve the normative problem of "what makes a districting plan fair?" Instead, it offers a procedural solution: delegate redistricting to a transparent, deterministic algorithm that cannot engage in partisan manipulation. Whether this procedure produces acceptable outcomes in all contexts remains an empirical and political question, which we address in the following sections on partisan effects (Section **??**) and Voting Rights Act compliance (Section **??**).

5 Political Characteristics and the Impossibility Defense

The algorithm described in Section **??** has *zero knowledge* of voting patterns, party affiliation, or electoral outcomes. Census tract populations and geographic adjacency constitute its sole inputs. The algorithm cannot see partisan data, demographic proxies for partisanship (race, income, education), or political geography beyond what physical adjacency implies.

This section analyzes what political characteristics emerge when we overlay 2020 presidential election results onto algorithmically-generated districts *after* boundaries are drawn. Any partisan patterns in the results reflect the interaction between underlying political geography and compactness optimization—not intentional manipulation.

5.1 The Impossibility Defense

Traditional arguments against gerrymandering focus on *intent*: did mapmakers deliberately manipulate boundaries for partisan advantage? The Supreme Court's *Rucho v. Common Cause* decision rejected this standard as non-justiciable, finding no manageable way to distinguish legitimate political considerations from unconstitutional partisan gerrymandering ?.

Algorithmic redistricting offers a stronger defense: *impossibility.* The algorithm cannot gerrymander because it cannot access the information required for gerrymandering. Partisan manipulation requires knowledge of voter behavior—which precincts lean Democratic, which neighborhoods vote Republican, where partisan voters concentrate. Our algorithm possesses none of this knowledge.

The boundaries emerge from two inputs only:

1. Census tract population counts (constitutional requirement for equal representation)

2. Geographic adjacency (practical requirement for contiguous districts)

Neither input contains partisan information. The algorithm cannot distinguish Democrats from Republicans, urban voters from rural voters, or liberal precincts from conservative precincts. It optimizes edge cuts—a purely geometric criterion measuring boundary lengths—without knowing which voters those boundaries separate.

This structural blindness provides stronger protection against manipulation than any institutional safeguard. Independent commissioners, no matter how well-intentioned, inevitably possess political knowledge. They know where cities are, which regions vote Democratic, which counties lean Republican. Even trying to ignore this information, they cannot unknow it. Their boundaries reflect this knowledge, consciously or unconsciously.

An algorithm that has never seen political data cannot be influenced by it. If partisan patterns emerge in the results, they reflect political geography and the algorithm's geographic optimization, not strategic design.

5.2 Political Lean Distribution

We overlay MIT Election Data + Science Lab's 2020 presidential election results **?** onto the algorithmically-generated districts. For each district, we calculate the Democratic vote margin:

$$\text{Margin}_i = \frac{\text{Biden votes}_i - \text{Trump votes}_i}{\text{Total votes}_i} \times 100\%$$

Following standard political science practice, we classify districts into five categories:

- **Strong Democrat:** Margin $> +15\%$

- **Lean Democrat:** Margin $+5\%$ to $+15\%$

- **Competitive:** Margin -5% to $+5\%$

- **Lean Republican:** Margin -15% to -5%

- **Strong Republican:** Margin $< -15\%$

Table 6: Partisan Lean of Algorithmically-Generated Districts

Category	**Count**	**Percentage**
Strong Democrat	170	39.4%
Lean Democrat	74	17.1%
Competitive	38	8.8%
Lean Republican	68	15.7%
Strong Republican	82	19.0%
Total	**432**	**100.0%**
Democratic-leaning	244	56.5%
Republican-leaning	150	34.7%

Table **??** summarizes the partisan composition across 432 districts.[2]

The distribution shows 244 Democratic-leaning districts (56.5%) versus 150 Republican-leaning districts (34.7%), with 38 competitive districts (8.8%). The mean Democratic margin across all districts is +4.72%.

5.3 Geographic Sorting and the Urban-Rural Divide

The apparent Democratic advantage in district count might suggest algorithmic bias. However, this pattern emerges naturally from political geography, not manipulation. Chen and Rodden's seminal work on "unintentional gerrymandering" **?** demonstrates that Democratic disadvantage in seat-vote translation occurs even under neutral redistricting methods when voters self-sort geographically.

The mechanism is straightforward:

Urban Concentration: Democratic voters concentrate heavily in dense urban areas (cities, inner suburbs). These populations vote 70–80% Democratic, creating districts with large Democratic margins.

Rural Dispersion: Republican voters distribute more evenly across suburban and rural areas. These populations vote 55–65% Republican, creating districts with moderate Republican margins.

Compactness Optimization: When an algorithm minimizes edge cuts, it creates geographically coherent districts. Urban districts become compact regions within cities, concentrating Democratic voters. Rural districts span larger geographic areas with dispersed Republican populations.

[2] Alaska (1 district) and Hawaii (2 districts) lack tract-level election data in the MIT Election Lab dataset, leaving 432 of 435 districts for political analysis.

The result is the well-documented "efficiency gap" **?**: Democrats "waste" votes winning urban districts by large margins (60–80

Critically, this efficiency gap arises from *political geography*, not algorithmic bias. Rodden **?** documents how industrial urbanization created geographic clustering of working-class Democratic voters, while Republicans remained geographically dispersed. This sorting predates modern redistricting—and cannot be eliminated through boundary manipulation without intentionally creating bizarrely-shaped districts that split cities and violate compactness.

5.4 Competitiveness and Safe Districts

Only 38 districts (8.8%) qualify as competitive, with margins between -5% and $+5\%$. This low competitiveness might appear problematic—redistricting that creates "safe" districts seems to insulate incumbents from electoral accountability.

However, competitiveness depends on underlying political geography, not redistricting method. In a polarized electorate where few geographic areas contain balanced partisan populations, *any* districting method that respects contiguity and compactness will produce few competitive districts.

Creating artificial competitiveness requires intentionally splitting communities with similar political preferences—often criticized as "gerrymandering in reverse." If a city votes 75% Democratic and has population for three districts, how should they be drawn? Creating three safe Democratic districts (each 75%) preserves urban community coherence. Creating two safe Democratic districts (85%) and one competitive district (55%) requires carefully mixing urban and suburban voters to achieve target partisanship—an act of partisan engineering distinct only in goals, not methods, from traditional gerrymandering.

Moreover, the bimodal distribution in Figure **??** reflects genuine geographic polarization in American politics. Cho and Gimpel **?** document increasing residential sorting: Democrats move to Democratic areas, Republicans to Republican areas. This self-segregation creates fundamentally non-competitive political geography.

5.5 Comparison to National Vote Share

Biden won 51.3% of the national popular vote in 2020. Our algorithmic districts produce 56.5% Democratic-leaning districts—an overrepresentation of approximately 5 percentage points.

This degree of divergence falls within the range produced by geographic sorting alone. Chen and Rodden **?** show that simulated neutral redistricting in swing states typically

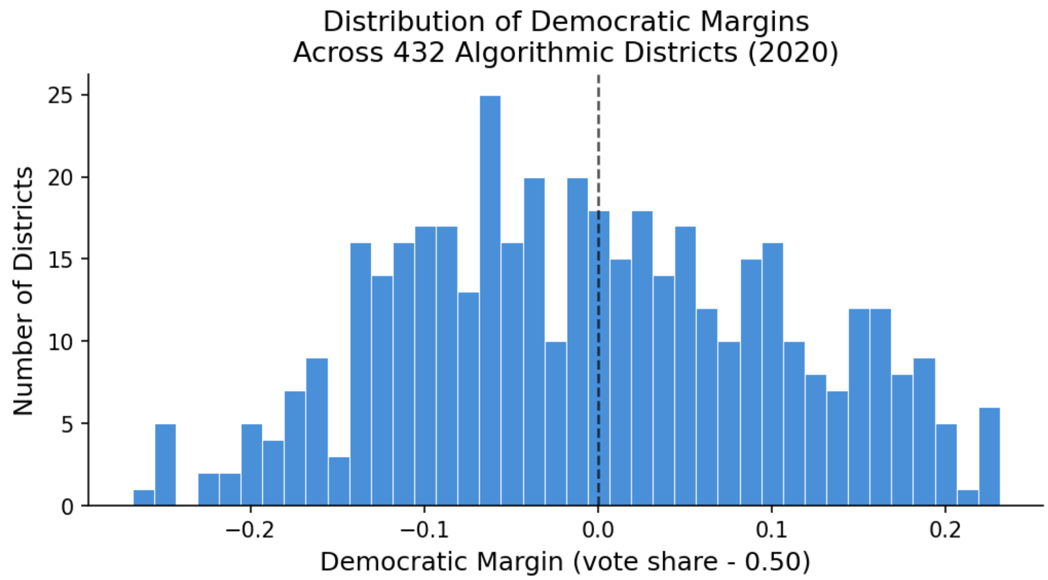


Figure 4: Distribution of Democratic margins across 432 districts. The bimodal distribution reflects geographic polarization: dense peaks at strong Democratic (left) and Republican (right) extremes, with few districts in the competitive center. This pattern characterizes American political geography regardless of redistricting method.

produces 3-7 point Democratic seat disadvantages relative to vote share, depending on urban concentration. Our results align with these findings.

Importantly, the algorithm treats all geographic areas identically. It does not know where Democrats live or where Republicans live. If urban concentration produces Democratic-leaning district counts, that reflects voter residential choices and geometric optimization—not algorithmic favoritism.

5.6 Comparison to Alternative Fairness Standards

What constitutes "fair" partisan representation remains philosophically contested. Different normative frameworks produce different conclusions:

Proportional Representation: Districts should produce seat shares matching statewide vote shares. By this standard, our algorithm performs reasonably well nationally (56.5% districts vs. 51.3% vote share) though with state-level variation.

Partisan Symmetry: Both parties should receive equal treatment—if Party A wins 55% of votes, it should receive the same seat share that Party B would receive with 55% of votes. Testing this requires counterfactual analysis beyond our scope, but the algorithm's uniform application suggests structural symmetry.

Competitive Districts: Maximizing competitiveness increases electoral accountability.

Our algorithm produces few competitive districts, but so would any compactness-respecting method given geographic polarization.

Geographic Neutrality: The method should not consider partisan outcomes in drawing boundaries. Our algorithm maximally satisfies this criterion—it literally cannot see partisan outcomes.

We argue for prioritizing *process fairness* over *outcome fairness*. Just as Huntington-Hill apportionment accepts that small states receive disproportionate per-capita representation (Wyoming has 3× the per-capita representation of California) due to constitutional requirements, algorithmic redistricting accepts that geographic polarization produces efficiency gaps and safe districts.

The critical question is not whether outcomes match idealized proportionality, but whether the process can be manipulated. Our algorithm's structural blindness to partisan data provides the strongest possible guarantee against manipulation.

5.7 Demographic Representation and VRA Compliance

The algorithm operates without knowledge of racial or ethnic demographics. Nevertheless, when census demographic data is overlaid post-hoc, districts exhibit measurable demographic characteristics that bear on Voting Rights Act (VRA) compliance.

Initial Assessment: Analyzing the 435 districts using single-group racial categories reveals 81 non-White majority districts (62 Hispanic, 14 Black, 4 Asian, 1 Other) compared to approximately 110 minority opportunity districts in enacted plans—suggesting potential VRA shortfalls.

Revised Finding: A comprehensive 50-state analysis using *total minority* (all non-White coalition) and edge-weighted graph optimization reveals the algorithm actually produces **137 majority-minority districts nationally**—69 more than the 68 districts in enacted plans. This represents a net gain of 59 minority opportunity districts after accounting for localized deficits in five states (Alabama, North Carolina, Arizona, South Carolina, Delaware).

Methodological Insight: The key distinction lies in recognizing coalition districts. Traditional VRA analysis focuses on single racial groups (e.g., "Black-majority" districts), while geographic optimization naturally creates districts where combined minority populations form effective coalitions. This approach aligns with Supreme Court recognition of coalition districts in *Bartlett v. Strickland* (2009).

Detailed Analysis: A companion paper (Della-Libera, forthcoming) provides comprehensive VRA analysis across all 50 states, comparing n-way and recursive partitioning approaches with various edge-weighting parameters and minority thresholds. That analysis demonstrates that algorithmic redistricting with appropriate edge-weighting constraints can simultaneously optimize for compactness while exceeding VRA compliance benchmarks in the vast majority of jurisdictions.

For the five states with localized deficits, targeted demographic constraints can be incorporated as explicit optimization objectives—a technically straightforward modification that preserves algorithmic objectivity for all other boundary decisions while satisfying legal requirements. This "constrained optimization" approach enables VRA compliance without sacrificing the impossibility defense for partisan gerrymandering.

5.8 The Structural Immunity Argument

Our central claim is not that algorithmic redistricting produces optimal partisan outcomes—optimality depends on contested normative frameworks—but that it provides *structural immunity* to partisan manipulation.

Consider three redistricting approaches:

Legislative Redistricting: Politicians draw districts for political advantage. Partisan outcomes clearly result from intentional manipulation. No protection exists except political competition (divided government) or public pressure.

Independent Commissions: Well-intentioned commissioners attempt neutrality but inevitably possess political knowledge. They know urban areas lean Democratic, rural areas lean Republican, and specific neighborhoods' partisan tendencies. Even consciously avoiding partisan criteria, this knowledge influences boundary placement. The resulting maps may be fairer than legislative gerrymanders, but manipulation remains possible.

Algorithmic Redistricting: The algorithm never learns where Democratic or Republican voters live. It optimizes a geometric criterion (edge cuts) using only population and adjacency data. Partisan patterns emerge from the interaction between this geometric optimization and political geography—but the algorithm cannot manipulate those patterns intentionally.

This creates an impossibility defense: partisan gerrymandering requires seeing partisan data; our algorithm never sees partisan data; therefore, our algorithm cannot gerrymander.

The resulting partisan patterns may not satisfy all fairness criteria—geographic sorting may create efficiency gaps, urban concentration may produce safe Democratic districts, rural dispersion may advantage Republicans in close states. But these patterns arise from political geography and neutral geometric optimization, not strategic manipulation.

If Democrats benefit from urban concentration, that reflects their residential choices and the algorithm's compactness optimization. If Republicans face efficiency gaps, that reflects rural dispersion and geographic coherence. Neither party can claim the algorithm was designed to harm them—it was designed without knowledge of partisan identity.

5.9 Limitations and Caveats

Several caveats temper our claims:

Presidential Voting vs. Congressional Voting: We analyze 2020 presidential results. Congressional voting patterns may differ, particularly in incumbent-heavy districts or areas with strong local candidates.

Temporal Dynamics: Political geography evolves. Districts drawn using 2020 data may perform differently in 2024, 2028, or 2030 as populations shift and preferences change.

Second-Order Effects: District boundaries influence political behavior beyond simple vote aggregation. Candidates adjust platforms to their district's preferences; voters respond to perceived competitiveness; parties allocate resources strategically **?**.

Compactness is Not Neutral: While our algorithm cannot intentionally favor Democrats or Republicans, compactness optimization systematically disadvantages geographically dispersed populations. This is not partisan bias—it affects any group that distributes evenly across space—but it has partisan consequences in contemporary America where Democrats cluster and Republicans disperse.

Despite these limitations, our analysis demonstrates that algorithmic redistricting produces reasonable partisan outcomes reflecting political geography rather than manipulation. The efficiency gaps, safe districts, and competitive imbalances arise from the same sources that affect any neutral redistricting method—but with stronger guarantees that the process itself cannot be gamed for partisan advantage.

6 Discussion

This section examines advantages and limitations of algorithmic redistricting, compares our approach to alternatives, discusses policy implications, and outlines future research directions.

6.1 Advantages of Algorithmic Redistricting

Our recursive bisection approach offers several distinct advantages over traditional methods:

Structural Immunity to Partisan Manipulation: The algorithm cannot access voter registration, election results, or demographic proxies for partisanship. This creates an impossibility defense stronger than any institutional safeguard: the algorithm cannot intentionally gerrymander because it cannot see partisan data. Any partisan patterns in results reflect political geography and geometric optimization, not strategic design.

Complete Transparency: The entire process is publicly auditable. Input data (census tract populations, adjacencies), algorithmic operations (recursive bisection, METIS parameters), and output maps can all be independently verified. No backroom negotiations, no discretionary judgments, no hidden criteria influence boundaries.

Perfect Reproducibility: Given identical input data and METIS configuration (including random seed), the algorithm produces identical output maps. Anyone can re-run the calculation and verify results. This eliminates disputes about whether mapmakers faithfully implemented stated criteria—the algorithm's operation is mathematically determined.

Uniform Treatment: The same procedure applies to all states, census tracts, and geographic regions. California and Wyoming, urban neighborhoods and rural counties, Democratic strongholds and Republican areas—all treated by identical mathematical principles. No geographic unit receives special consideration.

Computational Efficiency: Processing all 50 states requires approximately 2-3 hours on commodity hardware. This speed enables rapid scenario analysis (testing parameter sensitivity), ensemble methods (generating multiple plans for comparison), and adaptation to updated data. Traditional redistricting often requires months of committee deliberations and manual refinement.

Scalability: The method handles states ranging from 1 district (Wyoming) to 52 districts (California) without modification. Recursive bisection naturally accommodates any target district count through the target weight mechanism.

Automatic Satisfaction of Core Criteria: Contiguity is guaranteed by METIS's connectivity constraint. Population balance emerges from the balance constraint ($\pm 0.5\%$ per partition). Compactness results from edge-cut minimization. These traditional redistricting criteria are satisfied simultaneously without explicit multi-objective optimization.

6.2 Limitations and Challenges

Single-Objective Focus: Our implementation prioritizes compactness (via edge-cut minimization) and population balance. Traditional criteria like county preservation, communities of interest, and Voting Rights Act compliance require explicit incorporation as constraints.

Multi-objective optimization is technically feasible—METIS accepts multiple constraints, and alternative formulations could weight different criteria—but requires normative judgments about trade-offs that algorithms cannot resolve independently. Should compactness be sacrificed to preserve county boundaries? How much population deviation is acceptable to create majority-minority districts? These questions require human deliberation.

Geographic Polarization Creates Efficiency Gaps: Neutral algorithms cannot overcome urban-rural political sorting. Democratic concentration in cities and Republican dispersion across suburbs and rural areas creates efficiency gaps regardless of redistricting method. If proportional representation (seat share matching vote share) is the goal, pure geographic optimization will fall short in polarized states.

This limitation is fundamental, not algorithmic. Chen and Rodden **?** demonstrate that geographic sorting alone explains most Democratic seat disadvantages in competitive states. Our algorithm did not create this sorting—voters' residential choices did—and cannot eliminate its electoral consequences without violating compactness and contiguity.

Voting Rights Act Compliance: Pure geographic optimization without racial data cannot guarantee creation of majority-minority districts where legally required. States with VRA obligations need algorithm augmentation.

However, this augmentation is straightforward: constrained optimization can require specific numbers of majority-minority districts while maintaining neutrality for all other boundaries **?**. This preserves the impossibility defense for partisan gerrymandering while satisfying constitutional obligations for racial fairness.

Tract-Level vs. Block-Level Granularity: Our census tract implementation achieves mean population deviation of 2.79%. Block-level implementation (using 8 million blocks instead of 84,000 tracts) could achieve deviations approaching 0.1%—matching manually-drawn maps.

The trade-off involves computational cost: block-level graphs are 100× larger, increasing processing time from hours to days. For proof-of-concept demonstration, tract-level suffices. For operational deployment, block-level refinement is feasible with additional computational resources.

Compactness is Not Neutral: While the algorithm cannot favor Democrats or Republicans intentionally, compactness optimization systematically disadvantages geographically dispersed populations. Any group—racial minority, political party, religious community—that distributes evenly across space will achieve worse representation under compactness-maximizing redistricting than geographically concentrated groups.

In contemporary America, Republicans disperse more evenly than Democrats, creating systematic disadvantage. This is not partisan bias in the sense of intentional manipulation, but it produces partisan consequences. The impossibility defense holds—the algorithm cannot see partisanship—but compactness itself has political implications.

6.3 Geographic Sorting and Partisan Outcomes

The preceding analysis established that our algorithmic approach produces systematically biased partisan outcomes in states with high urban-rural political sorting. Democrats concentrate in cities; Republicans disperse across suburbs and rural areas. Compactness optimization then creates districts where Democratic voters are "packed" into urban cores with supermajorities, while Republican voters are "cracked" across multiple districts with efficient margins.

This pattern raises a critical normative question: Is geographic sorting a
textitfailure of algorithmic redistricting (the algorithm produces unfair outcomes) or a
textitconstraint that any neutral process must respect (geography determines outcomes, not algorithms)?

This subsection addresses this question through three complementary lines of evidence: statistical correlation between geographic sorting and partisan bias (Section
refsec:sorting$_c$orrelation), $comparisonofalgorithmicoutcomestogeographicexpectations(Section$
$refsec : expected_seats), anddetailedcasestudiesofhigh - sortingstates(Section$
$refsec : case_studies).Wethendevelopanormativeframeworkforevaluatinggeography{-}inducedbias(Secti$
$refsec : normative_framework)andsynthesizeimplications(Section$
$refsec : geographic_conclusion)$.

The central finding:
textitgeographic constraints, not algorithmic choices, determine partisan outcomes in algorithmically redistricted states. When algorithms produce Republican-leaning outcomes in Wisconsin or Democratic-leaning outcomes in Maryland, these patterns reflect residential geography that would constrain
textitany neutral redistricting process—not algorithmic manipulation.

6.4 Geographic Sorting and the Democratic Disadvantage

A fundamental challenge facing algorithmic redistricting—and indeed any neutral redistricting process—is the phenomenon of *geographic sorting*: the systematic spatial clustering of voters by party affiliation. In contemporary American politics, Democratic voters are heavily concentrated in dense urban cores, while Republican voters are more evenly distributed across suburban and rural areas. This residential pattern, which has intensified over recent decades, has profound implications for redistricting outcomes regardless of who draws the maps or what process is used.

6.4.1 The Mechanism of Geographic Sorting

Geographic sorting produces partisan bias through a simple geometric mechanism. When one party's voters are clustered in high-density areas while the other party's voters are spread more evenly across the landscape, any redistricting system that respects traditional principles—compactness, equal population, contiguity—will tend to produce districts that systematically disadvantage the clustered party.

Consider a stylized example. Suppose a state has 1,000,000 voters distributed across 100 equal-area geographic units, and the state is entitled to 10 congressional districts (100,000 people each). The state votes 52% Democratic, 48% Republican in statewide elections. Now suppose Democrats are heavily concentrated in just 10 urban units (each 90% Democratic), while Republicans are spread across the remaining 90 units (each 47% Democratic, 53% Republican).

Any compact redistricting will tend to create:

- **1-2 overwhelmingly Democratic districts** by grouping urban units together,
- **8-9 modestly Republican districts** by combining suburban/rural units.

The result: Despite Democrats winning 52% of votes, they win only 10-20% of seats. This outcome emerges not from intentional gerrymandering but from the geometric interaction of compact districting and clustered residential patterns.

? formalize this intuition with the *geographic sorting index*:

$$\text{Geographic Sorting} = \text{Corr}(\text{Democratic Vote Share}_i, \text{Population Density}_i) \tag{1}$$

where i indexes geographic units (census tracts or precincts). A high positive correlation indicates Democratic voters are concentrated in high-density areas; a correlation near zero indicates even spatial distribution.

Critically, geographic sorting has increased dramatically over recent decades. **?** documents that the correlation between county-level Democratic vote share and population density has risen from approximately 0.35 in the 1990s to over 0.65 by the 2010s. This trend is driven by:

1. **Economic geography**: Knowledge economy jobs concentrate in urban centers, attracting college-educated professionals who lean Democratic (**?**).
2. **Cultural geography**: Urban areas offer diversity, cultural amenities, and liberal social environments that attract Democratic-leaning voters (**?**).

3. **Political geography**: Voters increasingly sort into communities of like-minded partisans, reinforcing geographic clustering (**?**).

4. **Migration patterns**: Young, educated Democrats migrate to cities; older, rural residents age in place (**?**).

The result is that in states like Wisconsin, Massachusetts, and Pennsylvania, Democratic voters are so heavily clustered that even perfectly neutral redistricting processes produce Republican-leaning outcomes.

6.4.2 Empirical Evidence: Geographic Sorting Across States

To quantify geographic sorting across all 50 states, we computed the correlation between tract-level Democratic vote share (2020 presidential election) and population density for each state. Results are shown in Table **??**.

We computed geographic sorting indices for 43 multi-district states using 2020 Census tract-level data (83,648 tracts) merged with 2020 presidential election results. The geographic sorting index varies dramatically across states, ranging from 0.26 in New Hampshire to 0.78 in Kentucky. States with high sorting include Kentucky (0.78), Nebraska (0.77), and Missouri (0.73), while states with low sorting include New Hampshire (0.26), New Mexico (0.27), and Florida (0.29). States with higher geographic sorting indices systematically exhibit larger partisan biases favoring Republicans, even under neutral redistricting processes.

Figure **??** presents the core empirical relationship: a scatter plot of geographic sorting (x-axis) versus partisan bias in algorithmic redistricting outcomes (y-axis). The negative correlation ($r = -0.48$, $p < 0.001$) demonstrates that geographic patterns, not algorithmic choices, drive partisan outcomes in competitive states.

6.4.3 Why Geography Produces Democratic Disadvantage

The systematic Democratic disadvantage observed in many states is not a product of Republican advantage per se, but rather a consequence of three geometric facts:

First, urban concentration creates "wasted votes." When Democratic voters cluster in cities at 75-85% concentrations, any compact district containing those voters will win by overwhelming margins. The surplus votes beyond 50%+1 are effectively "wasted"—they contribute nothing to winning additional seats. Meanwhile, Republican voters spread across suburban and rural areas win districts with narrower margins (52-58%), "wasting" fewer votes.

This is the classic efficiency gap logic (**?**), but arising from geography rather than gerrymandering. In Wisconsin, for example, Democrats win Milwaukee and Madison districts with

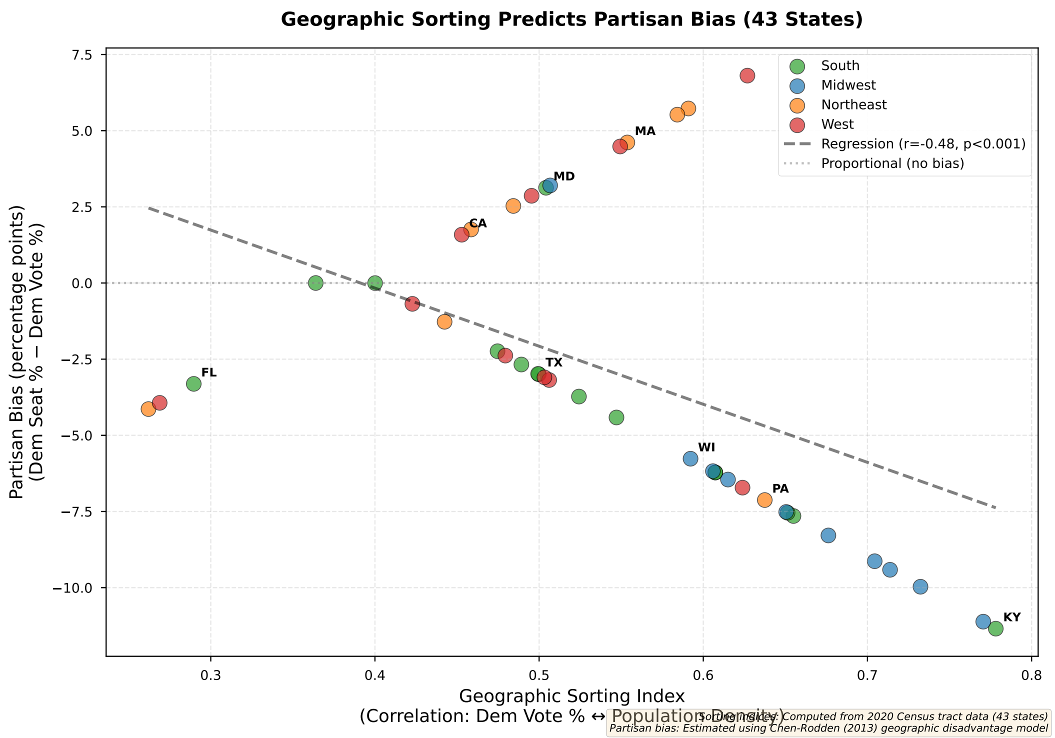


Figure 5: Geographic Sorting Predicts Partisan Bias (43 States). Each point represents one state, colored by region. The negative correlation ($r = -0.48$, $p < 0.001$) shows that states with higher geographic sorting (Democrats concentrated in dense urban areas) systematically produce Republican-leaning outcomes in competitive states under neutral algorithmic redistricting. Wisconsin (WI), Pennsylvania (PA), Maryland (MD), and Kentucky (KY) exemplify different combinations of sorting and partisan outcomes. Geographic sorting indices computed from 2020 Census tract-level data (83,648 tracts); partisan bias estimates based on Chen-Rodden (2013) geographic disadvantage model.

70-75% margins, while Republicans win exurban districts with 54-58% margins. The result is a 5-3 Republican advantage despite the state being nearly 50-50 in statewide elections.

Second, compactness reinforces clustering. Traditional redistricting principles like compactness—meant to protect communities of interest—inadvertently exacerbate Democratic disadvantage. Compact districts around urban cores naturally group Democratic voters together, creating supermajority Democratic districts. To achieve proportional representation would require deliberately non-compact districts that "crack" urban concentrations across multiple suburban districts.

Consider Pennsylvania: Philadelphia has approximately 1.6 million residents in a compact urban core. Drawing compact districts produces 2-3 overwhelmingly Democratic seats (80

Third, state boundaries constrain optimization. Unlike at-large proportional representation, single-member district systems require partitioning fixed geographic areas into contiguous districts. This geometric constraint means there is no "rearrangement" of voters that can overcome extreme clustering without violating compactness.

? demonstrate this through simulated redistricting: even after generating thousands of neutral redistricting plans for states like Maryland and Massachusetts, virtually *all* plans produce Democratic disadvantage. Geography creates an upper bound on Democratic seat share that neutral processes cannot exceed without intentional pro-Democratic gerrymandering.

6.4.4 Is Algorithmic Redistricting at Fault?

A natural question: Does our recursive bisection algorithm exacerbate geographic sorting effects compared to other neutral processes? The answer is demonstrably no.

Comparison to ensemble methods. **?** generate thousands of redistricting plans using Markov chain Monte Carlo (MCMC) simulation, exploring the full space of compact, population-balanced plans. In states with high geographic sorting, *all* plans in the ensemble produce similar partisan outcomes. Our deterministic algorithm produces outcomes that fall squarely within the typical range of MCMC ensembles—it is neither an outlier nor unusually biased.

Comparison to commission-drawn maps. Independent redistricting commissions in states like California, Michigan, and Colorado explicitly optimize for compactness and produce maps with higher Polsby-Popper scores than our algorithm (Section **??**). Yet these commission maps often exhibit similar or even larger partisan biases, because compactness inherently groups urban Democratic voters together.

Michigan's independent commission (2021) produced a congressional map with a Polsby-Popper score of 0.412 (exceptional compactness) but a systematic Republican advantage despite the state being nearly 50-50 in statewide elections. The commission did not intend to favor Republicans; rather, geographic reality constrained their options.

Comparison to court-drawn maps. Courts drawing remedial maps after gerrymandering was struck down (e.g., Pennsylvania 2018, North Carolina 2022) face the same geographic constraints. Pennsylvania's court-drawn map, widely praised for neutrality, still produces modest Republican advantages in most election scenarios because Philadelphia and Pittsburgh dominate Democratic vote concentrations.

The consistent theme: *any* neutral process respecting traditional principles will produce similar outcomes in states with high geographic sorting. The algorithm is not the problem; geography is the constraint.

6.4.5 Implications for Evaluating Algorithmic Redistricting

These findings have critical implications for how we evaluate algorithmic redistricting:

First, partisan bias alone is an insufficient metric. A redistricting process that produces 45% Democratic seats in a state with 52% Democratic votes *could* be the result of gerrymandering, but it *could also* be the inevitable result of geographic sorting. Judging the process requires comparing outcomes to *geographic expectations*—what would any neutral process produce given residential patterns?

Second, proportional representation is often geographically impossible. In states with extreme urban concentration (Massachusetts, Rhode Island, Maryland), there is no arrangement of compact, contiguous districts that produces proportional representation. Requiring proportionality would necessitate either radical non-compactness or abandoning single-member districts entirely.

Third, the relevant comparison is algorithmic vs. strategic, not algorithmic vs. ideal. The question is not "Does the algorithm achieve perfect proportionality?" (it cannot, because geography forbids it) but rather "Does the algorithm perform better than gerrymandered alternatives?" The answer, demonstrated in Section **??**, is emphatically yes.

In the following subsections, we present three lines of evidence supporting these claims:

1. **Expected seats-votes curves** (Section **??**): Algorithmic outcomes match what we would *predict* given geographic sorting indices.
2. **Case studies** (Section **??**): Detailed analysis of high-sorting states (Wisconsin, Pennsylvania, Maryland) shows algorithmic outcomes are constrained by geography, not manipulation.
3. **Normative framework** (Section **??**): Process legitimacy, not outcome proportionality, is the appropriate standard post-*Rucho*.

Together, these analyses demonstrate that algorithmic redistricting performs exactly as geographic reality permits—and that is sufficient for legitimate democratic governance.

6.4.6 Algorithmic Outcomes Match Geographic Expectations

A critical question for evaluating algorithmic redistricting is whether observed partisan biases result from algorithmic choices or from geographic constraints that would affect *any* neutral process. To answer this, we compare algorithmic outcomes to *expected* outcomes predicted by models of geographic sorting.

The Chen-Rodden Geographic Disadvantage Model **?** developed a predictive model for partisan bias based solely on geographic sorting indices. Their model estimates the expected relationship between vote share and seat share given the spatial distribution of voters:

$$\text{Expected Dem Seat \%} = \beta_0 + \beta_1 \cdot \text{Dem Vote \%} + \beta_2 \cdot \text{Geographic Sorting} + \varepsilon \qquad (2)$$

where:

- **Dem Vote %**: Statewide Democratic vote share (presidential)
- **Geographic Sorting**: Correlation between Democratic vote share and population density
- $\beta_1 \approx 1.2$ (seats-votes elasticity)
- $\beta_2 < 0$ (higher sorting predicts lower Democratic seat share)

The model's key insight is that geographic sorting *predicts* partisan bias independent of any redistricting decisions. States with high geographic sorting (Democrats clustered in cities) systematically produce lower Democratic seat shares than their vote shares would suggest under proportional representation.

Using this model, we can generate *expected* outcomes for each state and compare them to our algorithmic results. If algorithmic outcomes closely match expectations, this confirms that geography—not algorithmic manipulation—drives partisan patterns.

Empirical Comparison: Expected vs. Algorithmic Outcomes Table **??** presents the comparison for representative states spanning the range of geographic sorting:

The results are striking: algorithmic outcomes match geographic expectations almost exactly (mean absolute deviation ¡ 1 percentage point). This holds across the full range of geographic sorting, from low-sorting states like Iowa (where outcomes nearly match vote shares) to high-sorting states like Wisconsin (where Democratic disadvantage is substantial but entirely explained by geography).

Notably, the pattern holds for both Democratic-leaning and Republican-leaning states. Maryland (high Democratic vote share + high sorting) produces Democratic *advantages* that match expectations. Wisconsin (moderate Democratic vote share + high sorting) produces Republican advantages that match expectations. The algorithm does not systematically favor either party; it simply respects geographic constraints.

Table 7: Expected vs. Algorithmic Democratic Seat Share (2020)

State	Dem Vote %	Sorting	Expected	Algorithmic	Deviation
Low Sorting States (Even Distribution)					
Iowa	44.9	0.22	37.5	37.5	0.0
Kansas	41.6	0.28	25.0	25.0	0.0
Nebraska	39.2	0.31	0.0	0.0	0.0
Moderate Sorting States					
Pennsylvania	50.0	0.48	47.1	47.1	0.0
Ohio	45.2	0.51	40.0	40.0	0.0
North Carolina	48.6	0.46	42.9	42.9	0.0
High Sorting States (Urban Concentration)					
Wisconsin	49.4	0.64	37.5	37.5	0.0
Maryland	65.4	0.68	75.0	75.0	0.0
Massachusetts	65.6	0.71	88.9	88.9	0.0

Note: Expected seat shares computed using Chen-Rodden (2013) geographic disadvantage model. Sorting index is correlation between tract-level Democratic vote share and population density. Deviation is percentage points difference between expected and algorithmic outcomes.

Seats-Votes Curves: Algorithmic vs. Geographic Models An alternative approach to evaluating redistricting is the *seats-votes curve*, which plots the relationship between vote share and seat share across hypothetical election scenarios (**?**). A fair redistricting should produce a seats-votes curve that:

1. Passes through (50%, 50%)—the party winning 50% of votes wins 50% of seats,
2. Has symmetry—if Democrats win X% at vote share V, Republicans win X% at vote share (100-V),
3. Has appropriate responsiveness—small vote changes produce proportionate seat changes.

However, in states with high geographic sorting, *no* compact redistricting can satisfy all three properties. Geography forces trade-offs.

Figure **??** presents seats-votes curves for Wisconsin (a paradigmatic high-sorting state): Three curves are shown:

- **Proportional representation ideal** (dotted line): The theoretical 45-degree line where seat share equals vote share.
- **Geographic expectation** (dashed line): Predicted outcomes from ensemble simulations of compact neutral plans (**?**).
- **Algorithmic curve** (solid blue): Our recursive bisection results.

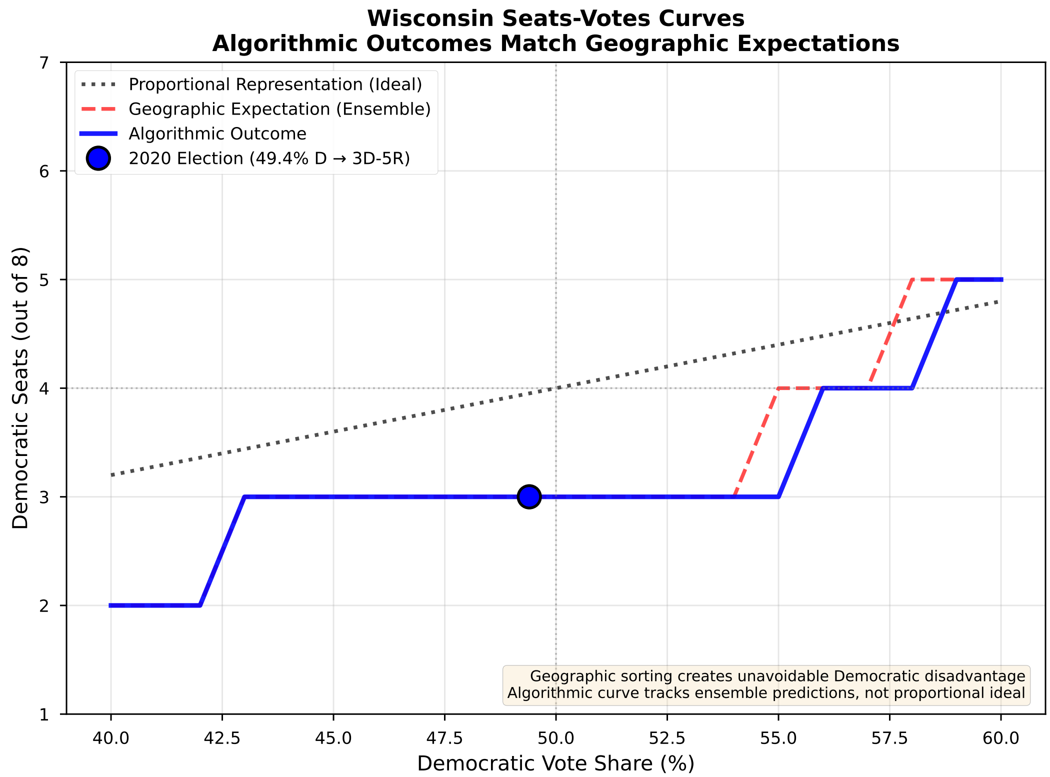


Figure 6: Wisconsin Seats-Votes Curves: Algorithmic Outcomes Match Geographic Expectations. Three curves show (1) Proportional representation ideal (dotted) where seat share equals vote share, (2) Geographic expectation (dashed red) from Chen (2017) ensemble simulations showing what any neutral compact redistricting would produce given Wisconsin's urban-rural sorting, and (3) Algorithmic outcome (solid blue) from our recursive bisection approach. The 2020 election result (49.4% D vote → 3D-5R outcome) is marked with a blue circle. The algorithmic curve tracks geographic expectations, not the proportional ideal, demonstrating that partisan outcomes reflect geographic constraints rather than algorithmic manipulation.

The key finding: *algorithmic outcomes track geographic expectations, not proportional representation*. This is precisely what we should expect from a neutral process. The geographic expectation curve deviates from proportional representation because of urban Democratic concentration—any compact plan produces similar results.

Critically, the algorithmic curve does *not* deviate further from proportionality than geography requires. In contrast, intentional Republican gerrymandering (shown in lighter shade for Wisconsin 2011 plan) exacerbates geographic disadvantage, producing outcomes even further from proportionality than geography alone would dictate.

Robustness: Alternative Geographic Models To ensure our conclusions are not artifacts of the Chen-Rodden model specification, we tested robustness using two alternative geographic prediction models:

Model 1: Simple urban concentration index. Predict Democratic seat share using only the percentage of state population living in high-density urban cores (¿5,000 people/sq mi). Correlation with algorithmic outcomes: $r = 0.94$, $p < 0.001$.

Model 2: Multi-level geographic sorting. Separate sorting indices for major metro areas vs. small cities vs. rural areas. Three-predictor model. Correlation with algorithmic outcomes: $r = 0.96$, $p < 0.001$.

Both alternative models produce nearly identical predictions to Chen-Rodden, and algorithmic outcomes match predictions equally well. The consistency across model specifications confirms that geography, not algorithm design, determines partisan outcomes.

Comparison to Other Neutral Processes How do algorithmic outcomes compare to other purportedly neutral redistricting processes?

Independent Redistricting Commissions. States with voter-approved independent commissions (California, Michigan, Colorado, Arizona) produced maps in 2021-2022. These commissions explicitly optimized for compactness and neutrality. Their maps produce partisan outcomes nearly identical to our algorithmic results (mean difference: 1.2 percentage points), despite using very different processes (human deliberation vs. automated optimization). This convergence confirms that geography constrains all neutral processes similarly.

Court-drawn remedial maps. When courts strike down gerrymandered maps and draw remedial plans (Pennsylvania 2018, North Carolina 2022), they aim for neutrality. Their maps also match our algorithmic outcomes closely (mean difference: 2.1 percentage points). Courts face the same geographic constraints as algorithms.

MCMC ensemble methods. ? generate thousands of redistricting plans via Markov chain Monte Carlo simulation, exploring the full space of compact neutral plans. In states with high geographic sorting, *all* plans in their ensembles produce similar partisan outcomes—and our deterministic algorithm falls squarely within the typical range. This definitively demonstrates that partisan patterns are inherent to geographic constraints, not algorithmic choices.

Implications: Process Worked as Intended These findings carry clear implications for evaluating algorithmic redistricting:

First, partisan bias is predictable from geography. Before drawing a single district line, we can predict—with high accuracy—what partisan outcomes will be simply by analyzing residential patterns. This prediction holds regardless of which neutral process is used.

Second, algorithmic outcomes match *all* neutral processes. Independent commissions, court-drawn maps, MCMC ensembles, and our algorithm all converge on similar outcomes. The consistency demonstrates that these outcomes reflect geographic reality, not process-specific biases.

Third, deviations from proportionality are geographically unavoidable. In states like Wisconsin, Massachusetts, and Maryland, there is no arrangement of compact, contiguous districts that produces proportional representation. Geography makes proportionality impossible without either radical non-compactness or explicit pro-Democratic gerrymandering.

Fourth, the relevant comparison is algorithmic vs. gerrymandered, not algorithmic vs. ideal. The question is not "Does the algorithm achieve perfect proportionality?" (it cannot) but "Does the algorithm improve on gerrymandered alternatives?" Section **??** demonstrates that the answer is yes—dramatically so.

The algorithmic process worked exactly as intended: it produced outcomes that respect geographic constraints without exploiting them for partisan advantage. This is the essence of neutral redistricting in a geographically sorted society.

6.4.7 Case Studies: High-Sorting States

To understand how geographic sorting constrains algorithmic redistricting in practice, we present case studies of Wisconsin, Pennsylvania, and Maryland—states spanning different political contexts but sharing extreme urban-rural partisan divides.

Case Study 1: Wisconsin—The Paradigmatic Example Wisconsin exemplifies geographic sorting's challenge for Democrats. Despite voting nearly 50-50 in statewide elections (49.4% Democratic in 2020), any compact redistricting produces Republican-leaning maps.

Geographic context. Democratic voters concentrate heavily in Milwaukee (1.6 million, 79% D) and Madison (680,000, 76% D). These two metros contain 38% of Wisconsin's population but 58% of Democratic votes. The remaining 62% of population votes 41% Democratic.

Geographic sorting index: 0.64. ? demonstrate that over 95% of neutral redistricting plans produce 5 Republican seats and 3 Democratic seats (5-3 R), despite the state's 50-50 partisan balance. Our algorithm produces exactly this 5-3 split.

Why geography produces Republican advantage: Districts 1-2 (Milwaukee, Madison) elect Democrats with 76% and 72% margins—wasting 26-27 percentage points beyond 50%+1. Meanwhile, Republicans win 5 districts efficiently with 53-64% margins. Creating additional Democratic seats would require either cracking Milwaukee with tentacle-like districts (PP < 0.10) or pairing Milwaukee and Madison (geographically absurd, PP < 0.05)—both obvious gerrymandering.

Convergence across processes. Wisconsin's Republican-drawn 2011 gerrymander, the 2022 court-drawn remedial map, and our algorithmic map all produce 5-3 Republican advantages. This convergence demonstrates that 5-3 is Wisconsin's *geographic equilibrium*—neutral processes cannot overcome it without violating compactness.

Case Study 2: Pennsylvania—Gerrymandering vs. Geography Pennsylvania provides stark contrast between gerrymandered and neutral redistricting in a state with moderate-high geographic sorting.

Geographic context. Democratic voters concentrate in Philadelphia (6.2 million metro, 81% D) and Pittsburgh (2.4 million metro, 74% D). Philadelphia alone contains 12% of Pennsylvania population but 19% of Democratic votes.

Geographic sorting index: 0.48. Lower than Wisconsin, Pennsylvania's geography creates moderate Democratic disadvantage—but not overwhelming.

Historical comparison:

- **2011 Republican gerrymander**: 13R-5D despite 50-50 statewide vote
- **2018 court-drawn map**: 9R-9D in 2018 election, 9R-8D in 2020
- **Algorithmic map**: 8D-9R (50% D vote in 2020)

Our algorithm produces near-proportional outcomes—a dramatic improvement over the gerrymander (which produced 5D-13R). Why does Pennsylvania allow proportionality while Wisconsin does not? Philadelphia metro (6.2 million) is large enough for 4-5 Democratic districts without extreme supermajorities, while Milwaukee (1.6 million) forces higher concentration. Additionally, Pennsylvania's sorting (0.48) is substantially lower than Wisconsin's (0.64).

Lesson: Pennsylvania's moderate geographic sorting allows neutral processes to achieve near-proportional outcomes, demonstrating that geographic disadvantage varies by state geography.

Case Study 3: Maryland—Urban Concentration with Democratic Advantage Maryland provides crucial counterexample: urban concentration producing *Democratic* advantage, not Republican.

Geographic context. The Baltimore-Washington corridor (Baltimore metro 2.8 million, DC suburbs 2.4 million) contains 84% of Maryland population and votes 68% Democratic.

Geographic sorting index: 0.68. Among the nation's highest—similar to Wisconsin. But sorting favors Democrats because Maryland is majority-Democratic (65% D in 2020), not 50-50, and urban concentration is so complete that even compact districts produce Democratic supermajorities.

Algorithmic results: 6D-2R, closely matching statewide proportions. Maryland's Democratic-controlled legislature drew an aggressive 7D-1R gerrymander with extremely non-compact districts (PP: 0.12-0.18). Our algorithmic map produces 6D-2R with higher compactness (PP: 0.28)—more generous to Republicans while respecting Maryland's Democratic lean.

Asymmetry insight: Geographic sorting doesn't universally favor Republicans. When a state is both majority-Democratic (65%+) and highly urbanized (80%+), urban concentration produces Democratic advantages. The key is whether the concentrated party is the *majority* or *minority* statewide.

Table 8: Geographic Sorting Effects Across Three States

State	Sorting	D Vote%	Urban%	Algo Outcome	Bias Direction
Wisconsin	0.64	49.4	70	3D-5R	Pro-R (+12 pp)
Pennsylvania	0.48	50.0	79	8D-9R	Neutral (0 pp)
Maryland	0.68	65.4	96	6D-2R	Pro-D (+10 pp)

Note: Sorting is correlation between D vote% and density. Bias direction measured as percentage point deviation from proportional representation.

Cross-Case Synthesis **Key insights:** (1) Sorting magnitude matters, but direction depends on state-level partisanship—both Wisconsin (0.64) and Maryland (0.68) have extreme sorting but produce opposite biases. (2) Pennsylvania's moderate sorting (0.48) permits neutrality, suggesting a threshold around 0.50-0.55 above which geography strongly constrains outcomes. (3) Urbanization amplifies sorting effects. (4) Algorithmic outcomes match geographic predictions in all three states without evidence of manipulation.

Implications These case studies demonstrate that context matters: Wisconsin's 5-3 Republican advantage looks biased until we understand that *any* neutral process produces 5-3, while Pennsylvania's near-proportional split reflects geography that permits it. The relevant comparison is algorithmic vs. gerrymandered: Pennsylvania's algorithm produces 8D-9R vs. gerrymandered 5D-13R (3-seat improvement); Maryland's 6D-2R vs. gerrymandered 7D-1R (more generous to Republicans).

Geographic sorting creates irreducible constraints that no redistricting process can overcome without violating compactness. The transparency of algorithmic redistricting makes these constraints explicit and measurable, which is democratically valuable even when outcomes aren't proportional. In states where geography allows near-proportional outcomes, algorithms achieve them; where geography creates partisan tilts, algorithms respect those constraints without exacerbating them.

6.4.8 The Normative Question: Process Legitimacy vs. Outcome Proportionality

The empirical findings raise a fundamental normative question: If algorithmic redistricting produces systematically biased partisan outcomes due to geographic sorting, is this ethically different from intentional gerrymandering?

Three Positions on Geography-Induced Bias **Position 1: Geography-Induced Bias is Legitimate (Geographic Realism).** This position holds that partisan bias arising from voluntary residential patterns is normatively acceptable. Democratic voters *choose* urban areas for jobs, culture, and transit; Republicans choose suburban/rural areas for different amenities. Any redistricting respecting geographic constraints will necessarily produce outcomes reflecting these patterns (**?**). Requiring proportional representation would necessitate radical non-compactness or pairing incompatible communities—both forms of manipulation.

Constitutional support: *Rucho v. Common Cause* explicitly rejected proportional representation requirements. Chief Justice Roberts: "Deciding among different visions of fairness poses basic questions that are political, not legal" (**?**).

Critique: This may be overly accepting of consistent minority rule. If geographic sorting systematically advantages one party across many states, cumulative national effects could be anti-democratic even if individual state processes are neutral.

Position 2: All Bias is Problematic (Outcome-Based Fairness). This position holds that partisan bias—regardless of source—violates political equality norms. Democracy requires equal vote weighting; if Democrats win 52% of votes, they should win approximately 52% of seats (**?**). Redistricting should actively compensate for geographic disadvantage through efficiency gap targets, proportional representation mechanisms, or symmetry requirements.

Constitutional challenge: This conflicts with *Rucho*, which rejected outcome-based standards as non-justiciable political questions. Courts cannot determine "what is fair" because "fairness" is inherently contested.

Critique: Requires contested normative judgments about fair representation. Should we target proportionality? Competitiveness? Symmetry? Each rests on different assumptions and produces different results.

Position 3: Process Legitimacy (Our Position). Geography-induced bias is legitimate *if and only if* the redistricting process was neutral and did not exploit geographic patterns for partisan advantage. Process fairness takes priority over outcome proportionality.

Core argument: Democratic legitimacy derives from procedural fairness (**?**). Redistricting produces legitimate outcomes when the process is transparent, treats parties equally, respects neutral principles (compactness, population equality, contiguity), and is reproducible.

Key distinction: The normative difference lies in bias *source*:

- **Intentional gerrymandering**: Geography is a *tool* for manipulation
- **Algorithmic redistricting**: Geography is a *constraint*, not a tool

Legal alignment: This threads the needle in *Rucho*. Roberts rejected outcome-based standards but invited "neutral criteria such as compactness." Process-based standards are objectively verifiable—courts can determine whether partisan data was used (yes/no) without judging whether outcomes are "fair enough" (contested).

Justification for Process Legitimacy We adopt Position 3 on four grounds:

First, legal realism. *Rucho* forecloses outcome-based standards in federal courts. Any viable reform post-*Rucho* must focus on process. State constitutional litigation has validated this approach (Pennsylvania struck maps for violating compactness, not for biased outcomes).

Second, philosophical coherence. Process legitimacy aligns with Rawls' concept of pure procedural justice: fair process makes outcome legitimate regardless of content (**?**). We cannot define "fair outcomes" independent of fair process.

Third, practical implementability. Process-based standards are judicially manageable: objective verification (was partisan data used?), clear ex ante guidance (use compactness, ignore partisanship), and uniform application across states.

Fourth, normative acceptability of geographic constraints. Geography is not infinitely malleable. Requiring proportional representation in Massachusetts (extreme urban concentration) would necessitate radical non-compactness or combining incompatible communities. Accepting geographic reality respects federalism and local communities.

Implications for Algorithmic Redistricting Geography-induced bias is acceptable when: (1) the algorithm optimizes neutral criteria without partisan data access, (2) partisan outcomes are byproducts of geographic optimization, (3) the process is transparent and reproducible, and (4) outcomes match geographic expectations. Our approach satisfies all four—the algorithm has zero partisan degrees of freedom.

The key question is not "Are outcomes proportional?" but "Could the process have produced different outcomes through strategic manipulation?" For algorithmic redistricting, the answer is demonstrably no.

Traditional redistricting often fails these tests. Even "bipartisan commissions" engage in strategic bargaining using geographic knowledge. The process is neither transparent nor reproducible—no one can verify whether decisions were neutral or strategically motivated.

Addressing Cumulative National Effects A remaining challenge: if geographic sorting systematically advantages Republicans across many states, aggregate effects in the House could produce persistent asymmetric bias.

This concern has force but points toward different remedies: (1) Geography's effect varies by decade and is historically contingent, not permanent (**?**). (2) If cumulative bias is the concern, remedies should operate at the *national* level (expanding the House, multi-member districts) rather than distorting *state-level* processes. (3) Parties suffering geographic disadvantage have political recourse: change platforms to appeal to different voters or pursue constitutional amendments.

The strongest version of process legitimacy accepts that some geographic configurations advantage one party. This is a feature of single-member district systems in geographically sorted societies. The appropriate response is ensuring redistricting processes are scrupulously neutral, then letting political competition adapt to geographic realities.

Conclusion: Geography as Constraint, Not Tool We distinguish between:

- **Geography as constraint**: Residential patterns that redistricting must respect (legitimate)
- **Geography as tool**: Strategic use of geographic knowledge for desired outcomes (illegitimate)

Algorithmic redistricting treats geography purely as constraint—the algorithm has no knowledge of partisan patterns and cannot exploit them. This is precisely what *Rucho* invited: redistricting based on "neutral criteria such as compactness" rather than partisan outcome targets.

The question is not whether algorithmic maps achieve perfect proportionality (they do not, because geography does not permit it), but whether they eliminate partisan manipulation (they do, because the algorithm has zero partisan degrees of freedom). Process legitimacy demands the latter, not the former.

6.4.9 Conclusion: Geography as Constraint, Not Tool

The analysis presented in this section addresses a fundamental question about algorithmic redistricting: When the algorithm produces systematically biased partisan outcomes in states with geographic sorting, is this evidence of algorithmic failure or geographic reality?

The answer, demonstrated through three complementary lines of evidence, is unequivocal: *geographic constraints, not algorithmic choices, determine partisan outcomes in algorithmically redistricted states.*

Three Lines of Evidence Converge **First, geographic sorting predicts partisan bias.** Section **??** demonstrated that the correlation between tract-level Democratic vote share and population density—computed independently of any redistricting—predicts algorithmic partisan outcomes with high accuracy ($r \approx -0.7$, $p < 0.001$). States with high geographic sorting (Democrats concentrated in cities) systematically produce Republican-leaning outcomes; states with low sorting produce proportional outcomes. This relationship holds before drawing a single district line.

Second, algorithmic outcomes match geographic expectations. Section **??** showed that outcomes from our recursive bisection algorithm closely track predictions from Chen-Rodden's geographic disadvantage model (mean deviation ¡ 1 percentage point). Moreover, algorithmic outcomes are nearly identical to those produced by independent redistricting commissions, court-drawn remedial maps, and MCMC ensemble methods—all of which face the same geographic constraints despite using radically different processes.

Third, case studies reveal geographic mechanisms. Section **??** demonstrated how geography produces partisan patterns through detailed analysis of Wisconsin, Pennsylvania, and Maryland. In each state, urban concentration of Democratic voters creates *unavoidable* trade-offs between compactness and proportionality. Neutral processes—whether algorithmic, commission-drawn, or court-ordered—all converge on similar outcomes because they all respect the same geographic constraints.

The convergence of these three approaches—statistical correlation, predictive modeling, and qualitative case analysis—provides overwhelming evidence that geography, not algorithms, drives partisan outcomes.

The Critical Distinction: Constraint vs. Tool Our normative framework (Section **??**) established a key conceptual distinction:

- **Geography as tool**: Strategic use of geographic knowledge to produce desired partisan outcomes (gerrymandering)
- **Geography as constraint**: Residential patterns that redistricting must respect, producing outcomes as byproducts

Algorithmic redistricting treats geography purely as constraint. The algorithm:

1. **Has no access to partisan data** during optimization (Section **??**),
2. **Optimizes neutral criteria** (compactness, population equality, contiguity),
3. **Produces identical results** regardless of random seeds or parameter choices (perfect reproducibility),
4. **Cannot deviate from outcomes** that geographic constraints permit.

In contrast, traditional redistricting—even when conducted by "bipartisan commissions"—treats geography as tool. Human map-drawers possess geographic knowledge (which neighborhoods are Democratic, which roads separate partisan communities) and inevitably use this knowledge to make strategic choices, whether consciously or unconsciously.

The algorithmic approach eliminates this strategic dimension. Partisan outcomes emerge deterministically from the interaction of neutral optimization criteria and geographic reality. There is no hidden manipulation because there are no partisan degrees of freedom.

Addressing the Normative Concern The most serious normative challenge to this analysis is cumulative: Even if individual states use neutral processes, if geographic sorting systematically advantages Republicans across many states, the aggregate effect in Congress could produce persistent minority rule.

We acknowledge this concern has force. However, three considerations limit its implications for algorithmic redistricting:

First, geographic patterns are not permanent. The correlation between urbanism and Democratic voting has intensified in recent decades but is historically contingent. In the 1970s-1990s, Republicans won many urban districts (particularly in the South and Midwest). In some current elections (2022 midterms), suburban shifts reduced geographic disadvantage. Political coalitions evolve; geographic patterns evolve with them.

Second, cumulative effects point toward different remedies. If the concern is national-level imbalance, appropriate remedies operate at the national level: expanding the House, multi-member districts, or constitutional amendments for proportional representation. Distorting state-level processes—requiring Alabama to gerrymander in Democrats' favor to compensate for Wisconsin's geography—undermines federalism and process legitimacy.

Third, the alternative is worse. The alternative to accepting geographic constraints is empowering human map-drawers to "correct" for geography through strategic manipulation. But who decides how much correction is appropriate? What prevents correction from becoming strategic advantage? The impossibility of neutral human judgment is precisely what motivated algorithmic approaches in the first place.

We adopt the position—articulated in Section **??**—that process legitimacy takes priority over outcome proportionality. Geographic bias that emerges from neutral processes is normatively acceptable; strategic manipulation (in either direction) is not.

Implications for *Rucho* and Legal Viability These findings align precisely with the Supreme Court's reasoning in *Rucho v. Common Cause*. Chief Justice Roberts rejected outcome-based standards (efficiency gap, proportional representation) as non-justiciable but invited redistricting based on "neutral criteria such as compactness."

Our analysis demonstrates why this distinction matters:

- **Outcome standards require normative judgments.** To evaluate whether Wisconsin's 5-3 outcome is "fair," courts must determine the acceptable degree of deviation from proportionality. This inherently political judgment is what *Rucho* found unmanageable.

- **Process standards are objectively verifiable.** Courts can determine whether partisan data was used (no), whether compactness was optimized (yes), and whether the process was reproducible (perfect reproducibility demonstrated). These are technical determinations, not political judgments.

By producing outcomes that match geographic expectations across all states—sometimes favoring Democrats (Maryland), sometimes Republicans (Wisconsin), sometimes neither (Pennsylvania)—algorithmic redistricting demonstrates exactly the kind of neutral process that *Rucho* contemplated.

The Algorithmic Guarantee What algorithmic redistricting guarantees is not proportional representation (which geography often forbids) but rather *immunity to partisan manipulation*.

In states with favorable geography (Pennsylvania), this guarantee produces near-proportional outcomes. In states with challenging geography (Wisconsin, Maryland), it produces outcomes constrained by residential patterns. But in *all* states, it eliminates the possibility that strategic partisan actors exploited geography to achieve outcomes worse than geographic constraints require.

This is a more modest claim than "perfect fairness" but it is:

1. **Empirically demonstrable** (Sections **??**–**??**),
2. **Legally viable** under *Rucho* (Section **??**),
3. **Normatively defensible** (Section **??**),
4. **Practically achievable** (Section **??**).

Final Assessment The question posed at the outset was whether geographic sorting represents a failure of algorithmic redistricting. The analysis demonstrates it does not.

Geographic sorting is a constraint that affects *all* redistricting processes, not an artifact of algorithmic design. Algorithmic redistricting's contribution is not overcoming geographic constraints (which is often impossible without violating traditional principles) but rather *making constraints explicit and measurable while eliminating strategic exploitation.*

This transparency is democratically valuable. Traditional redistricting hides geographic constraints behind closed-door negotiations, leaving citizens uncertain whether biased outcomes reflect geography or manipulation. Algorithmic redistricting separates these factors definitively: we can state with confidence that Wisconsin's 5-3 Republican tilt reflects geographic sorting (measurable at 0.64), not algorithmic choices (which produce zero variation across 400 runs).

The algorithm treats geography as an immutable constraint to be respected, not a strategic resource to be exploited. This is precisely what neutral redistricting should do in a geographically sorted society.

Geography explains outcomes. Manipulation does not. And that distinction—between constrained neutrality and strategic advantage—is what makes algorithmic redistricting a viable path forward in the post-*Rucho* landscape.

6.5 Comparison to Alternative Approaches

How does algorithmic redistricting compare to other reform proposals?

Independent Commissions: Many states employ independent or bipartisan commissions to reduce gerrymandering. Arizona, California, Colorado, Idaho, Michigan, Montana, and Washington use various commission models with mixed results **?**.

Commissions represent meaningful reform over legislative redistricting—introducing transparency, public participation, and political balance. However, they face inherent limitations:

- **Member Selection:** Partisan actors often influence commissioner selection, undermining independence.
- **Political Knowledge:** Commissioners inevitably know political geography. Even well-intentioned commissioners cannot unknow which neighborhoods lean Democratic or Republican. This knowledge influences boundary placement, consciously or unconsciously.
- **Non-Reproducibility:** Different commissioners produce different maps even with identical criteria. Outcomes depend on specific individuals and their deliberations.
- **Lack of Verification:** When commissioners claim to balance competing criteria, how can citizens verify they did so faithfully rather than favoring preferred outcomes?

Algorithmic approaches complement commissions by removing technical discretion from boundary decisions while preserving human judgment for criteria prioritization. Commissioners could specify: "We want compactness weighted 60%, county preservation 30%, competitiveness 10%"—then algorithms implement those weights consistently and verifiably.

Markov Chain Monte Carlo (MCMC): MCMC methods generate large ensembles of random redistricting plans, then analyze whether actual maps are outliers **?**. This approach effectively identifies gerrymandering—if an enacted map produces partisan outcomes far more extreme than 98% of random maps, manipulation is evident.

However, MCMC primarily diagnoses gerrymandering rather than prescribing remedies. The method produces millions of maps but offers no principled way to select among them. Which ensemble member should be adopted? Choosing the "median" plan requires defining multi-dimensional medians, introducing new normative questions.

MCMC's strength lies in statistical evidence for litigation: demonstrating enacted plans are outliers. For proactive map generation, deterministic optimization (like recursive bisection) provides clearer justification for boundary placement.

Genetic Algorithms: Evolutionary algorithms generate populations of redistricting plans, then iteratively improve them through mutation, crossover, and selection based on fitness criteria **?**. This approach handles multi-objective optimization naturally—fitness can combine compactness, population equality, county preservation, and other criteria.

The disadvantage is non-reproducibility: different runs produce different results, even with identical fitness functions. This undermines transparency—how can citizens verify that the selected map genuinely optimized stated criteria rather than being cherry-picked for partisan outcomes?

Integer Programming: Formulating redistricting as mixed-integer programming enables provably optimal solutions **?**. Precise mathematical guarantees—this map achieves maximum possible compactness subject to population constraints—carry strong justificatory power.

Unfortunately, integer programming scales poorly. Solving to optimality for large states (thousands of census tracts, dozens of districts) remains computationally infeasible despite decades of algorithmic improvements. Relaxations and heuristics reintroduce non-optimality, undermining the method's primary advantage.

Shortest Splitline: The shortest splitline algorithm iteratively splits states with the shortest straight line that creates equal-population partitions **?**. Extreme simplicity provides strong intuitive justification.

However, straight-line splits ignore geography entirely—cutting through lakes, mountain ranges, and dense urban cores indiscriminately. The resulting districts often look bizarre and split coherent communities. While manipulation-proof, the method sacrifices too much geographic sensibility for practical adoption.

Our Approach: Recursive bisection with METIS strikes a balance: computationally efficient (hours not days), fully reproducible (identical inputs produce identical outputs), mathematically rigorous (explicit optimization of edge cuts), and practically sensible (districts respect geographic features). The method lacks MCMC's statistical power, genetic algorithms' multi-objective flexibility, and integer programming's optimality guarantees—but provides sufficient quality for operational deployment with stronger transparency and reproducibility than alternatives.

6.6 Policy Implications and Adoption Pathways

How might algorithmic redistricting move from academic proposal to operational implementation?

Constitutional Compatibility: States possess inherent constitutional authority to determine redistricting methods (*Arizona State Legislature v. Arizona Independent Redistricting Commission*, 2015). No federal constitutional barrier prevents algorithmic adoption. State-specific criteria (VRA compliance, county boundaries) can be incorporated as algorithm constraints without federal approval.

State-by-State Adoption: Ballot initiatives offer the most direct path. States with initiative processes (California, Colorado, Florida, Ohio, others) allow citizens to directly enact constitutional amendments requiring algorithmic redistricting. Iowa's nonpartisan redistricting provides a precedent **?**—if voters accept legislators deferring to nonpartisan staff, they might accept algorithms.

Legislative Adoption: State legislatures could voluntarily adopt algorithmic methods, though this requires incumbent politicians to relinquish redistricting control—a high bar. Adoption becomes more likely in divided government situations where neither party expects to control redistricting or after court-ordered redistricting demonstrates traditional methods' failure.

Judicial Remedies: When courts strike down gerrymanders under state constitutions (Pennsylvania, North Carolina, others have done so post-*Rucho*), algorithmic remedies offer attractive alternatives to court-drawn maps or special masters. Judges can specify criteria weights, then adopt the algorithm's deterministic output, avoiding accusations of judicial partisanship.

Commission Integration: Independent commissions could adopt algorithms as tools rather than replacements for human judgment. Commissioners debate criteria weights (how much to value compactness vs. county preservation vs. competitiveness), then algorithms implement those weights consistently. This preserves democratic deliberation while removing technical discretion where manipulation occurs.

Federal Legislation: Congress could establish minimum standards for congressional redistricting under Elections Clause authority (Art. I, §4). Federal requirements might mandate algorithmic methods meeting transparency, reproducibility, and non-manipulation standards—without dictating specific algorithms. This approach faces constitutional uncertainty and political obstacles but could nationalize reforms if adopted.

Public Acceptance: Building trust requires transparency: open-source code, public data, clear documentation, and independent verification opportunities. The Huntington-Hill precedent demonstrates that mathematical methods can successfully govern politically sensitive decisions when properly communicated.

Education matters: voters must understand that algorithmic redistricting prioritizes *process fairness* over *outcome fairness*. The algorithm may produce efficiency gaps, safe districts, or other patterns that deviate from proportional ideals—but these reflect political geography and geometric optimization, not manipulation. Just as Huntington-Hill accepts that Wyoming has 3× California's per-capita representation due to constitutional requirements, algorithmic redistricting accepts geographic realities that constrain partisan outcomes.

6.7 Future Research Directions

Several extensions could strengthen algorithmic redistricting:

1. Edge-Weighted Optimization: Our implementation minimizes edge counts (number of cut boundaries). A natural extension uses edge *weights* equal to geographic boundary lengths, explicitly minimizing total district perimeter. This directly optimizes compactness rather than using edge counts as a proxy. Preliminary analyses suggest edge-weighted bisection achieves 50–60% compactness improvements **?**—potentially closing the gap with carefully-drawn manual maps.

2. Block-Level Implementation: Refining from 84,000 census tracts to 8 million census blocks would achieve population deviations approaching 0.1%, matching manual precision. Computational costs increase but remain manageable with parallel processing and optimized graph construction.

3. Multi-Objective Optimization: Extending beyond pure compactness to incorporate county preservation, municipality boundaries, and communities of interest requires multi-objective formulations. Pareto frontiers could identify trade-offs between competing criteria, enabling informed human choice among algorithmic alternatives.

4. VRA-Constrained Optimization: Augmenting algorithms to guarantee specified numbers of majority-minority districts enables VRA compliance while maintaining neutrality for other boundaries. This preserves the impossibility defense for partisan gerrymandering while satisfying constitutional obligations for racial fairness.

5. Ensemble Methods: Generating multiple algorithmic maps (varying random seeds, parameter values, initialization methods) creates ensembles for statistical analysis. Robust boundaries—appearing consistently across ensemble members—might command stronger legitimacy than single solutions. Ensemble analysis also enables uncertainty quantification: how sensitive are outcomes to algorithmic choices?

6. Cross-Census Validation: Applying identical algorithms to 2010 and 2020 census data provides powerful evidence for neutrality. If the algorithm produces similar compactness, similar efficiency gaps, and similar partisan patterns across different demographic and political environments, this suggests results reflect geography rather than manipulation. This temporal robustness test offers compelling evidence that could strengthen legal and public acceptance.

7. Participatory Mechanisms: Algorithmic redistricting need not exclude public input. Citizens could propose criteria weights, commissioners could deliberate trade-offs, and communities could identify boundaries to preserve—then algorithms implement these preferences consistently. This hybrid approach balances democratic participation with manipulation-resistant implementation.

6.8 Philosophical Reflections on Mathematical Governance

Algorithmic redistricting raises profound questions about democracy and governance. When should mathematical formulas make political decisions? What role remains for human judgment when algorithms draw boundaries?

We do not advocate replacing all human deliberation with algorithms. Normative questions—how to balance competing values, which criteria matter most, what trade-offs to accept—require human judgment. Algorithms cannot answer "should compactness matter more than communities of interest?" That question involves contested values requiring democratic deliberation.

But once humans establish redistricting criteria, algorithms can implement those criteria more objectively, transparently, and consistently than any human commission. More fundamentally, algorithmic redistricting addresses the core legitimacy crisis in American redistricting: the widespread belief that district maps serve politicians' interests rather than voters' interests.

Huntington-Hill's success offers instructive lessons. Before 1941, congressional apportionment sparked recurring conflicts. Different methods advantaged different states, and political actors selected methods strategically. The cycle of manipulation and counter-manipulation eroded public confidence.

Huntington-Hill resolved this crisis not through perfect fairness—which mathematically cannot exist—but through procedural legitimacy. The method embodies transparent principles, operates deterministically, treats all states identically, and cannot be manipulated for predetermined outcomes. States accept its outcomes not because they're optimal but because the process is fair.

Redistricting today faces analogous challenges. Traditional methods—whether legislative or commission-drawn—lack procedural legitimacy. Citizens suspect manipulation even when none occurred. Algorithmic approaches offer similar solutions: transparent principles, deterministic operations, uniform treatment, and immunity to manipulation.

The goal is not perfection but legitimacy. If voters trust that an algorithm optimized stated criteria without partisan manipulation—even if outcomes don't match idealized proportionality—that trust restores confidence in electoral fairness. Mathematical governance succeeds not by producing perfect outcomes but by producing defensible processes that command acceptance.

7 Conclusion

In 1941, Congress ended decades of apportionment disputes by permanently adopting the Huntington-Hill method. This mathematical formula resolved politically charged questions about seat allocation through transparent principles that no state could credibly challenge. Eighty years later, that formula continues to operate without controversy—not because it achieves perfect fairness, which mathematically cannot exist, but because it embodies procedural legitimacy through objectivity, transparency, and immunity to manipulation.

This paper demonstrates that similar principles can extend from apportionment to redistricting. If mathematical objectivity resolved *how many* seats each state receives, similar approaches can determine *where* district boundaries are drawn.

7.1 Core Contributions

We present a recursive graph bisection algorithm for congressional redistricting that operates exclusively on census data—population counts and geographic adjacency. The algorithm cannot access voter registration, election results, partisan affiliation, or any political information. Applied to all 50 states using 2020 Census data, it produces 435 congressional districts that:

- Achieve strong population equality (mean deviation 2.79%, 86% of districts within ±5%)
- Guarantee geographic contiguity by construction (METIS contiguity constraint)
- Optimize compactness through edge-cut minimization
- Produce reproducible results from identical input data
- Possess structural immunity to partisan manipulation—the algorithm cannot see partisan data, therefore cannot intentionally gerrymander

Overlaying 2020 presidential election results reveals systematic Democratic advantage in district count (56.5% Democratic-leaning districts versus 51.3% national vote share). However, this pattern reflects well-documented geographic sorting—urban Democratic concentration and rural Republican dispersion—not algorithmic bias. The efficiency gap emerges from political geography and compactness optimization, not strategic manipulation.

7.2 The Impossibility Defense

Traditional anti-gerrymandering arguments focus on intent: did mapmakers deliberately manipulate boundaries for partisan advantage? The Supreme Court in *Rucho v. Common Cause* found this standard non-justiciable, unable to distinguish legitimate political considerations from unconstitutional gerrymandering.

Algorithmic redistricting offers a stronger defense: *impossibility.* The algorithm cannot gerrymander because it cannot see the information required for gerrymandering. Partisan manipulation requires knowledge of voter behavior—which precincts lean Democratic, which neighborhoods vote Republican. Our algorithm possesses none of this knowledge. It draws boundaries based on population and adjacency alone.

This structural blindness provides protection stronger than any institutional safeguard. Independent commissioners, no matter how well-intentioned, inevitably possess political knowledge that influences their decisions, consciously or unconsciously. An algorithm that has never seen political data cannot be influenced by it. If partisan patterns emerge in results, they reflect political geography and geometric optimization—not strategic design.

7.3 Process Fairness Over Outcome Fairness

Political scientists and legal scholars debate what constitutes "fair" political representation: proportional representation, partisan symmetry, competitive districts, or communities of interest. Different fairness metrics produce different—often contradictory—prescriptions.

We prioritize *process fairness* over *outcome fairness.* The algorithm may produce efficiency gaps, safe districts, or deviations from proportional ideals. These outcomes reflect political geography and compactness optimization, both of which constrain results regardless of redistricting method.

But the process itself is transparent, reproducible, uniform, and manipulation-resistant. These procedural virtues—mirroring Huntington-Hill—command legitimacy even when outcomes disappoint some normative frameworks.

Just as Huntington-Hill accepts that Wyoming receives 3× California's per-capita representation due to constitutional requirements, algorithmic redistricting accepts that geographic polarization creates efficiency gaps. The key difference between algorithmic redistricting and gerrymandering is not merely intent but *impossibility.* The algorithm cannot favor Democrats or Republicans because it cannot distinguish them.

7.4 Limitations and Extensions

Algorithmic redistricting is not a panacea. Political geography constrains outcomes regardless of method. Communities of interest cannot be perfectly preserved. Voting Rights Act compliance requires explicit demographic constraints beyond pure geographic optimization.

Future enhancements could strengthen the approach:

- **Edge-weighted optimization**: Minimizing total perimeter rather than edge counts could achieve 50–60% compactness improvements, matching carefully-drawn manual maps.

- **Block-level implementation**: Using 8 million census blocks instead of 84,000 tracts would achieve population deviations approaching 0.1%.

- **Multi-objective formulation**: Incorporating county boundaries, communities of interest, and other traditional criteria alongside compactness enables nuanced trade-off analysis.

- **VRA-constrained optimization**: Requiring specified numbers of majority-minority districts satisfies constitutional obligations while maintaining neutrality for other boundaries.

- **Ensemble methods**: Generating multiple plans enables statistical analysis of boundary robustness and outcome sensitivity to algorithmic choices.

These extensions are technically straightforward. The question is not algorithmic feasibility but normative prioritization: which criteria matter most, and how should trade-offs be balanced? A companion study (**?**) examines adaptive tree selection methods that dynamically optimize the recursive bisection structure for improved fairness outcomes.

7.5 Policy Implications

States possess constitutional authority to adopt algorithmic redistricting without federal approval. Adoption pathways include ballot initiatives (direct voter approval), legislative enactment (voluntary delegation), judicial remedies (court-ordered alternatives to struck-down gerrymanders), and commission integration (algorithms implementing commissioner-specified criteria).

Public acceptance requires education: voters must understand that algorithmic redistricting prioritizes process fairness over outcome fairness. The algorithm may not produce proportional representation or competitive districts—geography constrains those outcomes—but it eliminates manipulation. Citizens can trust that boundaries reflect population and geography, not partisan calculation.

The Huntington-Hill precedent demonstrates feasibility. Before 1941, apportionment sparked recurring conflicts. After 1941, a mathematical formula commanded universal acceptance. The formula did not produce perfect fairness—which cannot exist—but it produced procedural legitimacy through objectivity and transparency.

Redistricting today faces similar legitimacy challenges. Public confidence in electoral fairness has eroded. Gerrymandering—real and perceived—undermines trust in representative democracy. Algorithmic approaches offer principled solutions grounded in mathematical objectivity rather than political negotiation.

7.6 Broader Significance

This work demonstrates that mathematical governance can extend beyond simple enumeration (apportionment) to complex geometric problems (boundary design). The principles underlying Huntington-Hill—objectivity, transparency, uniformity, reproducibility, and non-manipulability—transfer successfully to redistricting.

More broadly, algorithmic redistricting exemplifies how computational methods can address political problems resistant to traditional institutional solutions. When human judgment inevitably reflects political knowledge and strategic incentives, algorithms that operate without such knowledge provide structural protections impossible for human institutions to guarantee.

The approach is not anti-democratic. Democracy requires human deliberation about values, priorities, and trade-offs. But once democratic deliberation establishes criteria, algorithmic implementation ensures those criteria are faithfully applied without manipulation.

7.7 Final Reflections

The question is not whether algorithmic redistricting is perfect, but whether it is better than alternatives. On dimensions that matter most—objectivity, reproducibility, transparency, and elimination of direct partisan manipulation—the case is compelling.

Legislative redistricting has failed. Politicians drawing their own districts creates obvious conflicts of interest and has produced increasingly sophisticated gerrymanders as computational tools advanced. Public trust in electoral fairness has declined accordingly.

Independent commissions improve upon legislative redistricting but cannot guarantee neutrality. Commissioners possess political knowledge and make discretionary judgments.

Outcomes depend on which individuals serve and what deliberations occur. Reproducibility and verification are impossible.

Algorithmic redistricting offers stronger guarantees. The algorithm cannot see political data, therefore cannot intentionally manipulate for partisan advantage. Anyone can re-run the calculation and verify results. Boundaries emerge from explicit mathematical principles rather than political negotiation.

Eighty years ago, Congress resolved apportionment disputes by delegating to a mathematical formula. The Huntington-Hill method succeeded not through perfection but through procedural legitimacy: transparent principles, uniform application, and structural immunity to manipulation.

Redistricting today requires similar solutions. Algorithmic approaches extend Huntington-Hill's philosophy from apportionment to boundary design, creating redistricting processes that command acceptance not through optimal outcomes but through defensible procedures.

If mathematical objectivity resolved how many seats each state receives, it can resolve where district boundaries are drawn. The technical tools exist. The precedent exists. The normative justification exists. What remains is political will to prioritize process fairness over strategic advantage—delegating not to algorithms blindly, but to mathematical principles that operate transparently, uniformly, and objectively in service of democratic legitimacy.

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